Mathematical modelling of bacterial resistance to multiple antibiotics and immune system response

Chapter 1 — Introduction

1.1 Background and Motivation

Bacterial infections have historically been among the leading causes of morbidity and mortality worldwide. The discovery of antibiotics in the twentieth century represented one of the most significant advances in modern medicine, transforming once fatal infections into treatable conditions. However, the widespread and often indiscriminate use of antibiotics has led to the emergence of resistant bacterial strains, which now pose a serious threat to global public health.

Antibiotic resistance occurs when bacteria evolve mechanisms that allow them to survive exposure to drugs designed to kill or inhibit them. This phenomenon is not new — resistance has been observed shortly after the introduction of nearly every class of antibiotic ever developed. What has changed is the scale and speed at which resistance is spreading, combined with a slowdown in the development of new antimicrobial agents. The result is a growing mismatch between the infections clinicians face and the tools available to treat them.

Of particular concern are nosocomial infections — those acquired in hospital settings — where patients are already vulnerable and where resistant organisms circulate freely. These infections affect approximately one in ten hospitalised patients and are responsible for thousands of deaths and enormous healthcare costs annually. Understanding the dynamics of bacterial resistance is therefore not merely an academic exercise but a clinical and public health necessity.

Mathematical modelling offers a powerful approach to studying these dynamics. By translating biological interactions into equations, models allow researchers to analyse system behaviour, identify critical parameters, and evaluate treatment strategies in a controlled and reproducible way. This project applies such an approach, using a system of ordinary differential equations to

model the interaction between sensitive bacteria, resistant bacteria, immune cells, and multiple antibiotics within a single host.

1.2 Problem Statement

Despite decades of research, bacterial resistance to antibiotics continues to rise at an alarming rate. Clinical treatment decisions — such as which antibiotics to use, at what dose, and for how long — are often made without a full quantitative understanding of how resistance develops and how the immune system interacts with both the bacteria and the drugs. Most existing mathematical models either ignore the immune response entirely or treat it as a constant background effect, which fails to capture the dynamic and adaptive nature of host immunity.

Furthermore, the majority of models focus on either the spread of resistance across a population or the pharmacokinetics of a single antibiotic, rarely combining multiple antibiotics with a dynamic immune response in a single within-host framework. This leaves a gap in understanding how these factors interact simultaneously during the course of treatment in an individual patient.

This project addresses that gap by implementing and analysing the within-host model proposed by Daşbaşı and Öztürk (2016), which incorporates sensitive and resistant bacterial populations, a dynamic immune cell population, and the concentrations of multiple antibiotics simultaneously. The central question is: under what conditions does antibiotic treatment succeed in controlling infection, and what role does immune system strength play in determining that outcome?

1.3 Objectives of the Study

The objectives of this project are as follows:

- To implement the mathematical model of Daşbaşı and Öztürk (2016) numerically using MATLAB.
- To reproduce and verify the analytical equilibrium results through simulation.

- To investigate the effect of immune system strength on treatment outcome.
- To analyse how the fitness cost of resistance influences the persistence of resistant bacteria.
- To evaluate the consequences of early antibiotic termination on bacterial load.
- To compare single antibiotic therapy with combination therapy quantitatively.
- To perform a sensitivity analysis identifying the most critical model parameters.

1.4 Scope and Limitations

This study focuses exclusively on within-host dynamics — that is, the behaviour of bacteria, immune cells, and antibiotics within a single infected individual. It does not model the spread of infection between individuals or across a hospital population. The model assumes that resistance arises solely through spontaneous mutation during antibiotic exposure and does not account for horizontal gene transfer or conjugation between bacteria, which are known to be significant resistance mechanisms in clinical settings.

The immune response modelled here is specific and cell-mediated, represented by a single population of immune cells that grow logistically in proportion to bacterial load. The complexity of the real immune system, including innate immunity, antibody responses, and cytokine signalling, is not captured. All parameters are taken from published literature and are treated as fixed constants, so patient-to-patient variability is not considered. The model is also continuous and deterministic, meaning stochastic effects that may be important at low bacterial densities are not included.

1.5 Organization of the Report

This report is organised into six chapters. Chapter 2 provides a literature review covering bacterial resistance, the immune system's role in infection, multiple antibiotic therapy, and the use of mathematical models in infectious disease research. Chapter 3 presents the mathematical model in full, including the ODE system, parameter definitions, normalisation, and equilibrium analysis. Chapter 4 describes the numerical methods used, including the MATLAB implementation, solver settings, and model validation. Chapter 5 details the simulation setup for all cases and scenarios investigated. Chapter 6 presents and discusses all results, including time-course simulations, parametric studies, and sensitivity analysis. The report concludes with a summary of key findings and suggestions for future work.

Chapter 2 — Literature Review

2.1 Bacterial Infections and Antibiotic Resistance

Bacterial infections remain a leading cause of disease and death globally, affecting millions of people each year across all age groups and healthcare settings. The introduction of penicillin in the 1940s marked the beginning of the antibiotic era, offering effective treatment for infections that had previously carried high mortality. Since then, numerous classes of antibiotics have been developed, each targeting different aspects of bacterial physiology such as cell wall synthesis, protein production, or DNA replication.

However, bacteria are capable of rapid evolutionary adaptation. Resistance to antimicrobial agents is now recognised as both a logical and inevitable consequence of antibiotic use. Each new class of antibiotic introduced into clinical practice has been followed within years, and sometimes months, by the emergence of resistant strains. Resistance mechanisms include enzymatic inactivation of the drug, modification of the drug target, active efflux of the drug from the bacterial cell, and reduced membrane permeability. These mechanisms can arise through spontaneous chromosomal mutations or through the acquisition of resistance genes from other bacteria via horizontal gene transfer.

The clinical consequences of resistance are severe. Infections caused by resistant organisms are associated with longer hospital stays, higher treatment costs, greater risk of complications, and increased mortality. Multidrug-resistant organisms — bacteria resistant to three or more classes of antibiotics — are now regularly encountered in hospitals worldwide, and extensively drug-resistant and pan-resistant strains have been reported in several countries.

2.2 Role of the Immune System in Bacterial Infections

The immune system is the body's primary defence against infection. It comprises two broad arms: the innate immune system, which provides a rapid and non-specific first response to pathogens, and the adaptive immune system, which mounts a targeted and specific response based on the identity of the infecting organism. Both arms play important roles in controlling bacterial infections.

When bacteria enter the body, innate immune cells such as neutrophils and macrophages are rapidly recruited to the site of infection. These cells engulf and destroy bacteria through phagocytosis and the release of antimicrobial substances. If the innate response is insufficient to clear the infection, the adaptive immune system is activated. Specific immune cells, including T lymphocytes and B lymphocytes, are produced in response to bacterial antigens and work to eliminate the pathogen more precisely.

The strength and speed of the immune response varies considerably between individuals and is influenced by factors such as age, nutritional status, underlying disease, and immunosuppressive therapy. Patients with weakened immune systems — including those receiving chemotherapy, organ transplant recipients, and individuals with HIV — are at substantially higher risk of severe bacterial infections and are less able to clear them even with antibiotic support. This interaction between immune function and antibiotic efficacy is a central theme of the model studied in this project.

2.3 Multiple Antibiotic Therapy

The use of two or more antibiotics simultaneously, known as combination or multiple antibiotic therapy, has become standard practice for several types of bacterial infection. The rationale for this approach is threefold. First, different antibiotics act on different bacterial targets, so their effects can be additive or synergistic, producing greater bacterial killing than either drug alone. Second, combination therapy reduces the probability of resistance emerging, since a single mutation is unlikely to confer resistance to two drugs with different mechanisms simultaneously. Third, for mixed infections involving multiple bacterial species or strains with different susceptibility profiles, combinations ensure that all components of the infection are covered.

Multiple antibiotic therapy is particularly well established in the treatment of tuberculosis, caused by Mycobacterium tuberculosis, where the WHO recommends a combination of four drugs during the initial phase of treatment. Similar approaches are used for HIV-associated bacterial infections, hospital-acquired pneumonia, and complicated urinary tract infections. The trade-off is increased risk of side effects and greater selective pressure on the bacterial population, which can in some circumstances accelerate the emergence of multidrug resistance.

Mathematical models that incorporate multiple antibiotics simultaneously are better placed to evaluate these trade-offs and to optimise treatment protocols than single-drug models.

2.4 Mathematical Modelling in Infectious Diseases

Mathematical modelling has a long history in the study of infectious diseases. The foundational SIR model, which divides a population into susceptible, infected, and recovered compartments, was introduced in the early twentieth century and remains widely used today for studying epidemic dynamics. More sophisticated extensions of this framework have been developed to account for latent infection, vaccination, age structure, spatial heterogeneity, and many other biological complexities.

Within the field of antimicrobial resistance, mathematical models have been used to study the spread of resistant organisms within hospitals, to evaluate

antibiotic stewardship interventions, and to assess the impact of different treatment protocols on resistance prevalence at the population level. Ordinary differential equation systems are the most common modelling framework due to their tractability and the well-developed theory available for analysing their behaviour.

At the within-host level, models have been used to describe how bacterial populations grow and evolve under antibiotic pressure, how immune cells respond to infection, and how drug pharmacokinetics influence bacterial killing. These models are particularly valuable for informing dosing strategies and for understanding why some patients respond to treatment while others do not.

2.5 Within-Host Models vs Epidemiological Models

Mathematical models of infectious disease can be broadly divided into two categories based on their scale of analysis. Epidemiological models, also called between-host models, describe the spread of infection across a population of individuals. These models are used to predict outbreak trajectories, evaluate vaccination programmes, and assess the impact of public health interventions. The SIR model and its variants fall into this category.

Within-host models, by contrast, describe the dynamics of infection inside a single individual. These models track the populations of pathogens, immune cells, and drugs over time within one host and are used to understand the mechanisms of infection progression, immune evasion, and treatment response. Within-host models are particularly relevant for studying antibiotic resistance because resistance develops at the level of individual bacterial cells within a patient, driven by mutation and selection pressure from drugs and immunity.

The two types of model operate at different scales and answer different questions. Epidemiological models inform population-level policy while within-host models inform clinical treatment decisions. The model studied in this project is a within-host model, focused on the dynamics of bacteria, immune

cells, and antibiotics within one infected individual over the course of a 90-day treatment period.

2.6 Review of the Base Paper (Daşbaşı and Öztürk 2016)

The model implemented in this project is based on the work of Daşbaşı and Öztürk, published in SpringerPlus in 2016 under the title "Mathematical Modelling of Bacterial Resistance to Multiple Antibiotics and Immune System Response." This paper represents an important contribution to the within-host modelling literature because it simultaneously incorporates dynamic immune response and multiple antibiotic therapy, two features that are often treated separately or omitted entirely in prior models.

The authors construct a system of n+3 ordinary differential equations tracking sensitive bacteria, resistant bacteria, immune cells, and the concentrations of n antibiotics over time. Bacterial growth follows a logistic model with a shared carrying capacity. Resistant bacteria are assumed to carry a fitness cost modelled as a reduced growth rate. Resistance arises through spontaneous mutation of sensitive bacteria upon antibiotic exposure. Immune cells are recruited logistically in proportion to bacterial load, making the immune response dynamic and dependent on infection severity rather than fixed.

The paper identifies four equilibrium points: an infection-free equilibrium, two boundary equilibria, and an interior equilibrium where all populations coexist. Stability analysis using the Routh-Hurwitz criteria and Lyapunov functions shows that the system converges to either E_2 or E_3 depending on whether the threshold condition $A < B$ or $A > B$ holds, where A and B are composite parameters encoding the net reproductive success of sensitive and resistant bacteria respectively. Numerical simulations using parameters derived from tuberculosis treatment with isoniazid and pyrazinamide support the analytical findings. The paper concludes that immune system weakness is a key driver of persistent infection and that the model results align well with clinical observations.

Chapter 3 — Mathematical Model

3.1 Model Description and Assumptions

Model Description

This study examines the within-host dynamics of a bacterial infection in a single patient receiving multiple antibiotic treatment. Unlike epidemiological models such as SIR which track disease spread across a population, this model focuses on what happens inside one individual's body — specifically how sensitive bacteria, resistant bacteria, immune cells, and antibiotic concentrations interact and change over time.

The model is built as a system of ordinary differential equations (ODEs) with five state variables: the concentration of antibiotic-sensitive bacteria s(t), antibiotic-resistant bacteria r(t), host immune cells b(t), and the concentrations of two antibiotics — isoniazid INH denoted a1(t) and pyrazinamide PZA denoted a2(t). All five variables are normalized, meaning their values fall between 0 and 1, where 1 represents the maximum possible concentration or population size.

The central biological question the model addresses is: given a patient with a bacterial infection being treated with two antibiotics simultaneously, will the infection be cleared, will resistant bacteria take over, or will both types of bacteria persist alongside the immune response?

The answer depends critically on two threshold parameters A and B derived from the model parameters. Parameter A represents the net reproductive ability of sensitive bacteria after accounting for antibiotic killing and immune destruction. Parameter B represents the net reproductive ability of resistant bacteria after immune destruction. When A is less than B, only resistant bacteria survive in the long run. When A is greater than B, both sensitive and resistant bacteria coexist with immune cells at a stable equilibrium.

Assumptions

The following assumptions are made in constructing this model:

Assumption 1 — Logistic Growth of Bacteria Both sensitive and resistant bacteria grow logistically with a shared carrying capacity. This means bacterial growth slows down as the total population approaches its maximum sustainable size inside the host. This is more realistic than exponential growth because resources inside a host body are limited.

Assumption 2 — Resistance Arises Only Through Mutation Antibiotic resistance is assumed to develop solely through spontaneous genetic mutations in sensitive bacteria that are exposed to antibiotics. Other mechanisms of resistance acquisition such as horizontal gene transfer, plasmid exchange, or conjugation between bacteria are not included in this model.

Assumption 3 — Fitness Cost of Resistance Resistant bacteria are assumed to grow more slowly than sensitive bacteria. This reflects the well-documented biological reality that maintaining resistance genes carries a metabolic cost, reducing the competitive ability of resistant strains when antibiotics are absent.

Assumption 4 — Immune Cells Respond Proportionally to Bacteria Immune cells are recruited to the site of infection at a rate proportional to the current bacterial load. The carrying capacity of immune cells is modelled as a multiple of the total bacterial population. This captures the biological reality that the immune system ramps up its response specifically in response to the size of the infection.

Assumption 5 — Immune System Kills Both Bacterial Types Equally The immune system is assumed to kill sensitive and resistant bacteria at the same per capita rate η. No distinction is made between how the immune system recognizes or eliminates the two bacterial subtypes.

Assumption 6 — Antibiotics Affect Only Sensitive Bacteria Antibiotics kill only sensitive bacteria and induce mutations converting sensitive bacteria into resistant ones. Resistant bacteria, by definition, are unaffected by the antibiotics in the model.

Assumption 7 — Antibiotic Concentrations Decay at a Constant Rate Each antibiotic is supplied to the body at a constant rate and is cleared from the

body at a constant per capita rate. This leads to a simple first-order pharmacokinetic model where the antibiotic concentration approaches a steady state value determined by the ratio of supply rate to clearance rate.

Assumption 8 — No Reinfection From Outside The model considers only bacteria already present inside the host. There is no external source of new bacteria entering the system during the treatment period.

Assumption 9 — Homogeneous Mixing All populations — bacteria, immune cells, and antibiotics — are assumed to be uniformly distributed throughout the infection site. Spatial variation within the host is not considered.

Assumption 10 — Deterministic Model The model is fully deterministic, meaning the same initial conditions always produce the same outcome. Random effects such as stochastic mutation events or random fluctuations in immune response are not included.

No, this is **pure text and tables** — no MATLAB needed. Just write this in your report:

3.2 State Variables and Parameters

State Variables

The model consists of five state variables, all of which are normalized to dimensionless values in the range [0, 1]. Normalization is achieved by dividing each variable by its respective maximum capacity, which simplifies the mathematical analysis and ensures all variables remain biologically meaningful.

Variable	Meaning	Normalization	Range
s(t)	Population of antibiotic-sensitive bacteria at time t	S(t) divided by carrying capacity T	[0, 1]
r(t)	Population of antibiotic-resistant bacteria at time t	R(t) divided by carrying capacity T	[0, 1]

Variable	Meaning	Normalization	Range
b(t)	Population of host immune cells at time t	B(t) divided by ωT	[0, 1]
a1(t)	Concentration of INH (Isoniazid) at time t	A1(t) divided by δ1/μ1	[0, 1]
a2(t)	Concentration of PZA (Pyrazinamide) at time t	A2(t) divided by δ2/μ2	[0, 1]

Where T is the carrying capacity of bacteria, ω is the ratio of immune cell capacity to bacterial capacity, δi is the supply rate of the i-th antibiotic, and μi is the clearance rate of the i-th antibiotic.

Parameters

The model contains the following parameters. Each parameter is strictly positive as required for biological validity.

Parameter	Symbol	Meaning	Unit	Value Used
Sensitive bacteria growth rate	βS	Maximum per capita growth rate of sensitive bacteria	day^{-1}	0.8
Resistant bacteria growth rate	βR	Maximum per capita growth rate of resistant bacteria	day^{-1}	0.4
Fitness cost of resistance	c	Fractional reduction in growth rate due to carrying resistance genes	dimensionless	0.5
Immune killing rate	η	Rate at which immune cells kill bacteria per unit concentration	day^{-1}	0.3

Parameter	Symbol	Meaning	Unit	Value Used
Immune cell growth rate	k	Per capita proliferation rate of immune cells in response to infection	day^{-1}	0.6
Mutation rate — INH	α1	Rate at which sensitive bacteria mutate to resistant due to INH exposure	day^{-1}	0.02
Mutation rate — PZA	α2	Rate at which sensitive bacteria mutate to resistant due to PZA exposure	day^{-1}	0.06
Antibiotic killing rate — INH	d1	Rate at which INH kills sensitive bacteria	day^{-1}	0.15
Antibiotic killing rate — PZA	d2	Rate at which PZA kills sensitive bacteria	day^{-1}	0.35
INH clearance rate	μ1	Rate at which INH is cleared from the body	day^{-1}	0.06
PZA clearance rate	μ2	Rate at which PZA is cleared from the body	day^{-1}	0.03

Derived Threshold Parameters

These are not independent parameters but are calculated from the parameters above. They determine which equilibrium the system converges to.

Symbol	Formula	Biological Meaning
A	$(\beta S - \Sigma(\alpha i + di)) / (\beta S + \eta)$	Net survival fraction of sensitive bacteria after antibiotic and immune pressure

Symbol	Formula	Biological Meaning
B	$\beta R / (\beta R + \eta)$	Net survival fraction of resistant bacteria after immune pressure
C	$\Sigma\alpha i / (\beta R + \eta)$	Contribution of mutation from sensitive to resistant bacteria

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Relationship Between Parameters

$\beta R = (1 - c) \times \beta S$

meaning:

$c = 0 \rightarrow \beta R = \beta S$ (no fitness cost, resistance is free)

$c = 0.5 \rightarrow \beta R = 0.4$ (resistance costs 50% of growth rate)

$c = 1 \rightarrow \beta R = 0$ (resistance is lethal, bacteria cannot grow)

In this model $c = 0.5$ so $\beta R = (1 - 0.5) \times 0.8 = 0.4$

Notes on Units

All state variables are dimensionless after normalization. All rate parameters carry units of day^{-1} meaning they represent fractional changes per day. This is consistent with the clinical treatment timescale of the model which runs over 90 days representing a standard antibiotic treatment course.

3.3 The ODE System

The model is governed by a system of five ordinary differential equations describing the time evolution of sensitive bacteria, resistant bacteria, immune cells, and two antibiotic concentrations.

Equation 1 — Sensitive Bacteria

$ds/dt = \beta S \cdot s \cdot (1-s-r) - \eta \cdot s \cdot b - s \cdot [(\alpha 1+d1) \cdot a1 + (\alpha 2+d2) \cdot a2]$

Term	Expression	Meaning
Logistic growth	$\beta S \cdot s \cdot (1-s-r)$	Sensitive bacteria grow at rate βS but slow down as total bacteria s+r approach carrying capacity
Immune killing	$\eta \cdot s \cdot b$	Immune cells destroy sensitive bacteria at rate η proportional to both populations
Antibiotic effect	$s \cdot \Sigma(\alpha i+di) \cdot ai$	Antibiotics kill sensitive bacteria at rate di and convert them to resistant through mutation at rate αi

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Equation 2 — Resistant Bacteria

$dr/dt = \beta R \cdot r \cdot (1-s-r) - \eta \cdot r \cdot b + s \cdot [\alpha 1 \cdot a1 + \alpha 2 \cdot a2]$

Term	Expression	Meaning
Logistic growth	$\beta R \cdot r \cdot (1-s-r)$	Resistant bacteria grow at rate βR which is lower than βS due to fitness cost c
Immune killing	$\eta \cdot r \cdot b$	Immune cells kill resistant bacteria at the same rate η as sensitive bacteria
Mutation gain	$s \cdot \Sigma\alpha i \cdot ai$	Sensitive bacteria mutate into resistant ones upon antibiotic exposure — this is a gain for r and a loss for s

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Equation 3 — Immune Cells

$db/dt = k \cdot b \cdot (1 - b/(s+r))$

Term	Expression	Meaning
Logistic growth	$k \cdot b \cdot (1 - b/(s+r))$	Immune cells proliferate at rate k with carrying capacity equal to total bacterial load s+r. When bacteria decrease the immune response also contracts. When s+r = 0 immune cells decay at rate k

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Equation 4 and 5 — Antibiotic Concentrations

$da1/dt = \mu1 \cdot (1 - a1)$

$da2/dt = \mu2 \cdot (1 - a2)$

Term	Expression	Meaning
INH dynamics	$\mu1 \cdot (1 - a1)$	INH is supplied continuously and cleared at rate μ1 driving concentration toward normalized steady state of 1
PZA dynamics	$\mu2 \cdot (1 - a2)$	Same structure as INH with its own clearance rate μ2

Complete System

$ds/dt = \beta S \cdot s \cdot (1-s-r) - \eta \cdot s \cdot b - s \cdot [(\alpha1+d1) \cdot a1 + (\alpha2+d2) \cdot a2]$

$dr/dt = \beta R \cdot r \cdot (1-s-r) - \eta \cdot r \cdot b + s \cdot [\alpha1 \cdot a1 + \alpha2 \cdot a2]$

$db/dt = k \cdot b \cdot (1 - b/(s+r))$

$da1/dt = \mu1 \cdot (1 - a1)$

$da2/dt = \mu2 \cdot (1 - a2)$

Biological Constraints

The following constraints must hold at all times for the system to remain biologically valid:

Constraint	Meaning
$s(t) \geq 0$	Sensitive bacteria cannot be negative
$r(t) \geq 0$	Resistant bacteria cannot be negative
$0 \leq b(t) \leq s+r$	Immune cells are bounded by bacterial load
$0 \leq ai(t) \leq 1$	Antibiotic concentrations are normalized
$s(t) + r(t) \leq 1$	Total bacteria cannot exceed carrying capacity

3.4 Normalization of Variables

Why Normalize

The original model variables operate on very different scales. Bacterial populations S(t) and R(t) are measured in number of cells and can reach values as large as 10^9 bacteria. Immune cell populations B(t) operate on a different scale entirely, and antibiotic concentrations Ai(t) are measured in mg/kg. Working with variables of such different magnitudes makes both mathematical analysis and numerical simulation unnecessarily complicated.

Normalization rescales all variables to dimensionless quantities between 0 and 1 by dividing each variable by its natural maximum value. This simplifies the algebra, reduces the number of independent parameters, and makes the system easier to analyse and interpret.

Original Unnormalized System

The original system before normalization is:

$$dS/dt = \beta S \cdot S \cdot (1 - (S+R)/T) - \lambda \cdot S \cdot B - S \cdot \Sigma\, \bar{\alpha}i \cdot Ai - S \cdot \Sigma\, \bar{d}i \cdot Ai$$

$$dR/dt = \beta R \cdot R \cdot (1 - (S+R)/T) - \lambda \cdot R \cdot B + S \cdot \Sigma\, \bar{\alpha}i \cdot Ai$$

$$dB/dt = k \cdot B \cdot (1 - B/(\omega(S+R)))$$

$$dAi/dt = \delta i - \mu i \cdot Ai \text{ for } i = 1, 2$$

Where:

Symbol Meaning

$S(t)$	Sensitive bacteria population (cells)
$R(t)$	Resistant bacteria population (cells)
$B(t)$	Immune cell population (cells)
$A_i(t)$	Concentration of i-th antibiotic (mg/kg)
T	Carrying capacity of bacteria (maximum population)
λ	Per capita immune killing rate
ω	Ratio of immune cell capacity to bacterial capacity
δ_i	Constant supply rate of i-th antibiotic
μ_i	Clearance rate of i-th antibiotic
$\bar{\alpha}_i$	Mutation rate due to i-th antibiotic
$\bar{d}_i$	Death rate of sensitive bacteria due to i-th antibiotic

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Substitution Step

We introduce the following normalized variables:

$$s(t) = \frac{S(t)}{T} \qquad r(t) = \frac{R(t)}{T} \qquad b(t) = \frac{B(t)}{\omega T}$$

$$a_i(t) = \frac{A_i(t)}{\delta_i/\mu_i}$$ where $\frac{\delta_i}{\mu_i}$ is the steady state of A_i

Rearranging to express original variables in terms of normalized ones:

$S = s \cdot T$

$R = r \cdot T$

$B = b \cdot \omega \cdot T$

$Ai = ai \cdot (\delta i/\mu i)$

Deriving Each Normalized Equation

Derivation of ds/dt

Start with the original dS/dt equation and substitute $S = sT$, $R = rT$, $B = b\omega T$, $Ai = ai(\delta i/\mu i)$:

$dS/dt = \beta S \cdot S \cdot (1 - (S+R)/T) - \lambda \cdot S \cdot B - S \cdot \Sigma\, \bar{\alpha}i \cdot Ai - S \cdot \Sigma\, \bar{d}i \cdot Ai$

Substitute:

$d(sT)/dt = \beta S \cdot (sT) \cdot (1-(sT+rT)/T) - \lambda \cdot (sT) \cdot (b\omega T) - (sT) \cdot \Sigma\, \bar{\alpha}i \cdot ai \cdot (\delta i/\mu i) - (sT) \cdot \Sigma\, \bar{d}i \cdot ai \cdot (\delta i/\mu i)$

Simplify term by term:

Term 1: $\beta S \cdot (sT) \cdot (1-(sT+rT)/T)$

$= \beta S \cdot (sT) \cdot (1 - s - r)$

Term 2: $\lambda \cdot (sT) \cdot (b\omega T)$

$= \lambda \cdot \omega \cdot T^2 \cdot s \cdot b$

→ divide through by T: $\lambda \cdot \omega \cdot T \cdot s \cdot b$

→ define $\eta = \lambda \cdot \omega \cdot T$

Term 3: $(sT)\cdot\Sigma\, \bar{\alpha}i\cdot ai\cdot(\delta i/\mu i)$

→ divide through by T: $s \cdot \Sigma\, \bar{\alpha}i\cdot(\delta i/\mu i)\cdot ai$

→ define $\alpha i = \bar{\alpha}i\cdot(\delta i/\mu i)$

Term 4: $(sT)\cdot\Sigma\, \bar{d}i\cdot ai\cdot(\delta i/\mu i)$

→ divide through by T: $s \cdot \Sigma\, \bar{d}i\cdot(\delta i/\mu i)\cdot ai$

→ define $di = \bar{d}i\cdot(\delta i/\mu i)$

Since $S = sT$ we have $dS/dt = T \cdot ds/dt$. Dividing both sides by T:

$ds/dt = \beta S \cdot s \cdot (1-s-r) - \eta \cdot s \cdot b - s \cdot \Sigma(\alpha i + di) \cdot ai$

Derivation of dr/dt

Start with original dR/dt and substitute $R = rT$, $S = sT$, $B = b\omega T$, $Ai = ai(\delta i/\mu i)$:

$dR/dt = \beta R\cdot(rT)\cdot(1-(sT+rT)/T) - \lambda\cdot(rT)\cdot(b\omega T) + (sT)\cdot\Sigma\, \bar{\alpha}i\cdot ai\cdot(\delta i/\mu i)$

Simplify term by term:

Term 1: $\beta R\cdot(rT)\cdot(1 - s - r)$

Term 2: $\lambda\cdot(rT)\cdot(b\omega T)$

$= \lambda\cdot\omega\cdot T^2 \cdot r \cdot b$

→ divide by T: $\eta \cdot r \cdot b$ (using $\eta = \lambda\cdot\omega\cdot T$)

Term 3: $(sT) \cdot \Sigma\, \bar{\alpha}i\cdot(\delta i/\mu i)\cdot ai$

→ divide by T: $s \cdot \Sigma\, \alpha i \cdot ai$

Dividing both sides by T:

$dr/dt = \beta R \cdot r \cdot (1-s-r) - \eta \cdot r \cdot b + s \cdot \Sigma\, \alpha i \cdot ai$

Derivation of db/dt

Start with original dB/dt and substitute $B = b\omega T$, $S = sT$, $R = rT$:

$dB/dt = k \cdot B \cdot (1 - B/(\omega(S+R)))$

$= k \cdot (b\omega T) \cdot (1 - (b\omega T)/(\omega(sT+rT)))$

$= k \cdot (b\omega T) \cdot (1 - (b\omega T)/(\omega T(s+r)))$

$= k \cdot (b\omega T) \cdot (1 - b/(s+r))$

Since $B = b\omega T$ we have $dB/dt = \omega T \cdot db/dt$. Dividing both sides by ωT:

$db/dt = k \cdot b \cdot (1 - b/(s+r))$

Derivation of dai/dt

Start with original dAi/dt and substitute $Ai = ai \cdot (\delta i/\mu i)$:

$dAi/dt = \delta i - \mu i \cdot Ai$

$d(ai \cdot \delta i/\mu i)/dt = \delta i - \mu i \cdot ai \cdot (\delta i/\mu i)$

$(\delta i/\mu i) \cdot dai/dt = \delta i - \delta i \cdot ai$

$(\delta i/\mu i) \cdot dai/dt = \delta i \cdot (1 - ai)$

Dividing both sides by $(\delta i/\mu i)$:

$dai/dt = \mu i \cdot (1 - ai)$

New Parameters After Normalization

The normalization step reduces the number of independent parameters by combining them:

New Parameter	Formula	Meaning
η	$\lambda \cdot \omega \cdot T$	Combined immune killing rate in normalized system
αi	$\bar{\alpha} i \cdot (\delta i/\mu i)$	Normalized mutation rate for i-th antibiotic
di	$\bar{d} i \cdot (\delta i/\mu i)$	Normalized killing rate for i-th antibiotic
βR	$(1-c) \cdot \beta S$	Resistant growth rate incorporating fitness cost

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Final Normalized System

After substitution and simplification the complete normalized system is:

$$ds/dt = \beta S \cdot s \cdot (1-s-r) - \eta \cdot s \cdot b - s \cdot [(\alpha 1+d1)\cdot a1 + (\alpha 2+d2)\cdot a2]$$

$$dr/dt = \beta R \cdot r \cdot (1-s-r) - \eta \cdot r \cdot b + s \cdot [\alpha 1\cdot a1 + \alpha 2\cdot a2]$$

$$db/dt = k \cdot b \cdot (1 - b/(s+r))$$

$$da1/dt = \mu 1 \cdot (1 - a1)$$

$$da2/dt = \mu 2 \cdot (1 - a2)$$

With all variables satisfying:

$s, r \geq 0$

$0 \leq b \leq s + r \leq 1$

$0 \leq ai \leq 1 \qquad \text{for } i = 1, 2$

This normalized system is mathematically cleaner, has fewer parameters, and all variables are bounded between 0 and 1, making both analytical and numerical work significantly more tractable.

3.5 Biological Feasibility Region

What Is the Feasibility Region

For a mathematical model to be biologically meaningful, all state variables must remain non-negative and bounded for all time $t \geq 0$. It would make no biological sense for a bacterial population to become negative or for a normalized concentration to exceed 1. The biological feasibility region Ω is the set of all values of the state variables that are biologically valid.

We define Ω and then prove that any solution starting inside Ω stays inside Ω for all future time. This property is called positive invariance and it guarantees that the model never produces biologically impossible values.

Definition of the Feasibility Region

The biological feasibility region is defined as:

$$\Omega = \{ (s, r, b, a1, a2) \in R^5 :$$

$$s \geq 0,$$

$$r \geq 0,$$

$$0 \leq b \leq s + r \leq 1,$$

$$0 \leq a1 \leq 1,$$

$$0 \leq a2 \leq 1 \quad \}$$

In plain English this means:

Constraint	Biological Meaning
$s \geq 0$	Sensitive bacteria population cannot be negative
$r \geq 0$	Resistant bacteria population cannot be negative
$s + r \leq 1$	Total bacteria cannot exceed normalized carrying capacity
$0 \leq b \leq s+r$	Immune cells cannot exceed total bacterial load and cannot be negative
$0 \leq a1 \leq 1$	INH concentration is normalized between 0 and 1
$0 \leq a2 \leq 1$	PZA concentration is normalized between 0 and 1

Proposition — Positive Invariance of Ω

Statement: The region Ω defined above is positively invariant with respect to the normalized system. That is, any solution $(s(t), r(t), b(t), a1(t), a2(t))$ that starts in Ω at time $t = 0$ remains in Ω for all $t \geq 0$.

Proof:

We prove positive invariance by examining each constraint separately and showing that the vector field of the system on the boundary of Ω always points inward or is tangent to the boundary — never outward.

Step 1 — Prove that s + r stays between 0 and 1

Add the first two equations of the normalized system:

$$ds/dt + dr/dt = \beta S \cdot s \cdot (1-s-r) - \eta \cdot s \cdot b - s \cdot \Sigma(\alpha i + di) \cdot ai$$

$$+ \beta R \cdot r \cdot (1-s-r) - \eta \cdot r \cdot b + s \cdot \Sigma \alpha i \cdot ai$$

Collecting terms:

$$d(s+r)/dt = \beta S \cdot s \cdot (1-s-r) + \beta R \cdot r \cdot (1-s-r) - \eta \cdot b \cdot (s+r) - s \cdot \Sigma di \cdot ai$$

Since all parameters are positive and $b \geq 0$ and $ai \geq 0$, the last two terms are non-negative subtractions. Therefore:

$d(s+r)/dt \leq \beta S \cdot s \cdot (1-s-r) + \beta R \cdot r \cdot (1-s-r)$

$= (\beta S \cdot s + \beta R \cdot r) \cdot (1-s-r)$

$\leq \beta S \cdot (s+r) \cdot (1-s-r)$ since $\beta R < \beta S$

Let $u = s + r$. Then:

$du/dt \leq \beta S \cdot u \cdot (1 - u)$

This is a logistic inequality. Its solution satisfies:

If $0 \leq u(0) \leq 1$ then $0 \leq u(t) \leq 1$ for all $t \geq 0$

Interpretation: Total bacteria s+r starts between 0 and 1 and can never exceed 1 or go below 0. The carrying capacity is never violated.

Step 2 — Prove that a1 and a2 stay between 0 and 1

The antibiotic equations are independent of all other variables:

$da1/dt = \mu 1 \cdot (1 - a1)$

$da2/dt = \mu 2 \cdot (1 - a2)$

Solving these first order linear ODEs directly:

$a1(t) = 1 - (1 - a1(0)) \cdot e^{(-\mu 1 \cdot t)}$

$a2(t) = 1 - (1 - a2(0)) \cdot e^{(-\mu 2 \cdot t)}$

Checking the bounds:

If $0 \leq a1(0) \leq 1$:

When $t \to 0$: a1(0) stays at initial value

When $t \to \infty$: $a1(t) \to 1$ (approaches but never exceeds 1)

Since $e^{(-\mu 1 \cdot t)} \in (0,1]$ for all $t \geq 0$:

$(1 - a1(0)) \cdot e^{(-\mu 1 \cdot t)} \geq 0$

therefore $a1(t) \leq 1$

Also $(1 - a1(0)) \cdot e^{(-\mu 1 \cdot t)} \leq (1 - a1(0)) \leq 1$

therefore $a1(t) \geq 0$

Same argument applies to a2(t).

Interpretation: Both antibiotic concentrations start between 0 and 1, increase monotonically toward their steady state of 1, and never leave the interval [0,1].

Step 3 — Prove that b stays between 0 and s+r

The immune cell equation is:

$db/dt = k \cdot b \cdot (1 - b/(s+r))$

First we check the lower bound. If $b = 0$ then:

$db/dt = k \cdot 0 \cdot (1 - 0/(s+r)) = 0$

The derivative is zero when $b = 0$, meaning the solution cannot cross from $b > 0$ to $b < 0$. So if $b(0) \geq 0$ then $b(t) \geq 0$ for all $t \geq 0$.

Now we check the upper bound. Let $u = s + r$ be fixed. The equation for b is logistic with carrying capacity $u = s+r$. If $b = s+r$ then:

$db/dt = k \cdot b \cdot (1 - b/b) = k \cdot b \cdot 0 = 0$

The derivative is zero at the boundary $b = s+r$. Furthermore if $b > s+r$ then:

$1 - b/(s+r) < 0 \rightarrow db/dt < 0$

So b is forced back down toward s+r whenever it tries to exceed it. Combined with the lower bound result:

If $0 \leq b(0) \leq s(0)+r(0)$ then $0 \leq b(t) \leq s(t)+r(t)$ for all $t \geq 0$

Interpretation: Immune cells are always bounded above by the total bacterial population. When bacteria are eliminated the immune response collapses as well. When bacteria grow the immune response can grow with them up to the same level.

Summary of the Proof

Constraint	Method Used	Conclusion
$s + r \leq 1$	Logistic inequality comparison	Total bacteria bounded by carrying capacity for all t
$s + r \geq 0$	Logistic inequality	Total bacteria never negative
$0 \leq ai \leq 1$	Exact solution of linear ODE	Antibiotics converge monotonically to 1
$b \geq 0$	Vector field at boundary b=0	Zero derivative means b cannot go negative
$b \leq s+r$	Vector field at boundary b=s+r	Zero derivative at boundary, negative above it

---

Conclusion

Since all five constraints are preserved by the dynamics of the system, the region Ω is positively invariant. This means every solution that begins inside Ω remains inside Ω for all $t \geq 0$. All state variables remain biologically meaningful throughout the entire simulation. This result justifies the mathematical analysis of equilibria and stability that follows in Chapter 4, since all equilibrium points we find will necessarily lie within Ω.

3.6 Threshold Parameters A, B and C

Definition

Three threshold parameters are derived from the model parameters. These parameters determine which equilibrium the system converges to and therefore determine the long term outcome of the infection.

Parameter	**Formula**
A	$(\beta S - \Sigma(\alpha i + di)) / (\beta S + \eta)$
B	$\beta R / (\beta R + \eta)$
C	$\Sigma\alpha i / (\beta R + \eta)$

Written explicitly for two antibiotics:

$$A = \frac{\beta S - (\alpha 1+d1) - (\alpha 2+d2)}{\beta S + \eta}$$

$$B = \frac{\beta R}{\beta R + \eta}$$

$$C = \frac{\alpha 1 + \alpha 2}{\beta R + \eta}$$

Biological Meaning of Each Parameter

Parameter A — Net Survival Ability of Sensitive Bacteria

$$\beta S - (\alpha 1+d1) - (\alpha 2+d2)$$

$$A = \frac{\beta S - \Sigma(\alpha i + di)}{\beta S + \eta}$$

Numerator: $\beta S - \Sigma(\alpha i + di)$

↑ ↑

how fast how much is lost

sensitive to antibiotics

bacteria (killing + mutation)

grow

Denominator: $\beta S + \eta$

↑ ↑

growth immune

rate pressure

A represents the fraction of sensitive bacteria that survive after accounting for antibiotic killing, mutation losses, and immune system pressure. A higher value of A means sensitive bacteria are more likely to persist. A lower value means antibiotics and immune cells together are successfully suppressing sensitive bacteria.

Parameter B — Net Survival Ability of Resistant Bacteria

$$B = \frac{\beta R}{\beta R + \eta}$$

Numerator: βR

↑

how fast resistant

bacteria grow

(already reduced

by fitness cost c)

Denominator: $\beta R + \eta$

↑ ↑

growth immune

rate pressure

B represents the fraction of resistant bacteria that survive after immune system pressure. Note that antibiotics do not appear in B because resistant bacteria are by definition unaffected by antibiotics. B is always strictly between 0 and 1 since all parameters are positive.

Parameter C — Mutation Contribution

$$C = \frac{\alpha_1 + \alpha_2}{\beta R + \eta}$$

Numerator: $\alpha_1 + \alpha_2$

↑

total mutation

rate from both

antibiotics

Denominator: $\beta R + \eta$

↑ ↑

resistant immune

growth pressure

C represents the relative contribution of mutation from sensitive to resistant bacteria compared to the forces acting against resistant bacteria. A higher C means antibiotics are generating more resistant mutants relative to how quickly the immune system can eliminate them.

The Key Threshold Condition

The entire long term behaviour of the system depends on comparing A and B:

Condition	Outcome
$A < B$	Sensitive bacteria die out. Only resistant bacteria and immune cells persist at equilibrium E2. Antibiotics have failed to prevent resistance dominance.
$A > B$	Both sensitive and resistant bacteria coexist with immune cells at equilibrium E3. Neither type is eliminated.

$A < B$ → E2 is stable → resistant bacteria dominate

$A > B$ → E3 is stable → coexistence of both types

Numerical Verification for Both Cases

Case I — A < B

Parameters: $\beta S=0.8$, $\beta R=0.4$, $\eta=0.3$, $\alpha 1=0.02$, $d1=0.15$, $\alpha 2=0.06$, $d2=0.35$

$$A = (0.8 - (0.02+0.15) - (0.06+0.35)) / (0.8 + 0.3)$$

$$= (0.8 - 0.17 - 0.41) / 1.1$$

$= 0.22 / 1.1$

$= 0.2000$

$B = 0.4 / (0.4 + 0.3)$

$= 0.4 / 0.7$

$= 0.5714$

$A = 0.2000 < B = 0.5714 \rightarrow$ E2 is stable

Case II — A > B

Parameters: $\beta S=0.8$, $\beta R=0.4$, $\eta=0.3$, $\alpha 1=0.06$, $d1=0.35$, $\alpha 2=0.02$, $d2=0.15$

$A = (0.8 - (0.06+0.35) - (0.02+0.15)) / (0.8 + 0.3)$

$= (0.8 - 0.41 - 0.17) / 1.1$

$= 0.22 / 1.1$

$= 0.2000$

$B = 0.4 / (0.4 + 0.3)$

$= 0.5714$

Wait — both cases give $A = 0.2000 < B = 0.5714$ with these parameters. To achieve $A > B$ you need stronger sensitive bacteria growth or weaker antibiotics. Use:

Parameters: $\beta S=0.8$, $\beta R=0.1$, $\eta=0.05$, $\alpha 1=0.001$, $d1=0.01$, $\alpha 2=0.001$, $d2=0.01$

$A = (0.8 - 0.011 - 0.011) / (0.8 + 0.05)$

$= 0.778 / 0.85$

= 0.9153

B = 0.1 / (0.1 + 0.05)

= 0.1 / 0.15

= 0.6667

A = 0.9153 > B = 0.6667 → E3 is stable

Case	A	B	Condition	Stable Equilibrium
Case I	0.2000	0.5714	A < B	E2 — resistant bacteria dominate
Case II	0.9153	0.6667	A > B	E3 — coexistence

Chapter 4 — Mathematical Analysis

4.1 Equilibrium Points

Finding Equilibrium Points

Equilibrium points are found by setting all derivatives equal to zero and solving the resulting algebraic system.

ds/dt = 0

dr/dt = 0

db/dt = 0

dai/dt = 0 for i = 1, 2

Step 1 — Solve Antibiotic Equations First

From $dai/dt = 0$:

$\mu i \cdot (1 - ai) = 0$

$ai = 1 \quad \text{for } i = 1, 2$

All equilibrium points have $a1 = a2 = 1$. Substituting $ai = 1$ into the remaining three equations reduces the system to:

$\beta S \cdot s \cdot (1-s-r) - \eta \cdot s \cdot b - s \cdot \Sigma(\alpha i + di) = 0 \quad \text{... (i)}$

$\beta R \cdot r \cdot (1-s-r) - \eta \cdot r \cdot b + s \cdot \Sigma \alpha i = 0 \quad \text{... (ii)}$

$k \cdot b \cdot (1 - b/(s+r)) = 0 \quad \text{... (iii)}$

Step 2 — Solve Immune Cell Equation

From equation (iii) either:

$b = 0$ (no immune response)

OR

$b = s + r$ (immune cells at maximum capacity)

These two cases are solved separately.

Case 1 — When b = 0

Substituting $b = 0$ into (i) and (ii):

$s \cdot [\beta S \cdot (1-s-r) - \Sigma(\alpha i + di)] = 0$

$r \cdot [\beta R \cdot (1-s-r)] = 0$

This gives either $s = 0$ or the bracket equals zero, and either $r = 0$ or the bracket equals zero.

Subcase $s = 0, r = 0$ → gives E0

Subcase $s = 0, r \neq 0$ → from $\beta R\cdot(1-r) = 0$ → $r = 1$ → gives E1

Subcase $s \neq 0, r \neq 0$ → gives a point where s and r have opposite signs → biologically meaningless → rejected

---

Case 2 — When $b = s + r$

Substituting $b = s+r$ into (i) and (ii):

$$s \cdot [\beta S\cdot(1-s-r) - \eta\cdot(s+r) - \Sigma(\alpha i+di)] = 0 \quad \text{... (iv)}$$

$$r \cdot [\beta R\cdot(1-s-r) - \eta\cdot(s+r)] + s\cdot\Sigma\alpha i = 0 \quad \text{... (v)}$$

Subcase $s = 0$ → substituting into (v):

$$r \cdot [\beta R\cdot(1-r) - \eta\cdot r] = 0$$

$$r \cdot [\beta R - (\beta R+\eta)\cdot r] = 0$$

$r = \beta R/(\beta R+\eta) = B$ → gives E2

Subcase $s \neq 0$ → from (iv) the bracket equals zero:

$$\beta S\cdot(1-s-r) - \eta\cdot(s+r) - \Sigma(\alpha i+di) = 0$$

$$(\beta S+\eta)\cdot(1-s-r) = \beta S - \Sigma(\alpha i+di)$$

$$(s+r) = [\beta S - \Sigma(\alpha i+di)]/(\beta S+\eta) = A = \bar{b}$$

Substituting back gives $\bar{s}$ and $\bar{r}$ → gives E3

The Four Equilibrium Points

Using the definitions A, B, C from Section 3.6:

$$A = \frac{\beta S - \Sigma(\alpha i+di)}{\beta S + \eta} \qquad B = \frac{\beta R}{\beta R+\eta} \qquad C = \frac{\Sigma\alpha i}{\beta R+\eta}$$

Equilibrium	Values	Name
E0	(0, 0, 0, 1, 1)	Infection free
E1	(0, 1, 0, 1, 1)	Resistant bacteria only
E2	(0, B, B, 1, 1)	Resistant bacteria and immune cells
E3	($\bar{s}$, $\bar{r}$, $\bar{b}$, 1, 1)	Coexistence of all populations

Where the E3 components are:

$$\bar{s} = \frac{A(A-B)}{A-B+C} \qquad \bar{r} = \frac{A \cdot C}{A-B+C} \qquad \bar{b} = A$$

Existence Conditions

Equilibrium	Exists When	Biological Meaning
E0	Always	System can always be infection free in theory
E1	Always	Full resistant bacteria dominance always possible
E2	Always	Resistant bacteria and immune coexistence always possible
E3	A > B only	Coexistence only when sensitive bacteria are relatively strong

---

Key Observation

E0 and E1 always exist but as shown in Chapter 4.2 they are always unstable — the system never stays at these points. The system always moves toward either E2 or E3 depending on whether A < B or A > B.

4.2 Local Stability Analysis

What Local Stability Means

A equilibrium point is locally asymptotically stable (LAS) if solutions that start close to it return to it over time. In biological terms: if the system is slightly disturbed from equilibrium, does it recover back to that equilibrium or move away from it?

LAS: disturb → system returns → stable

Unstable: disturb → system moves away → unstable

The Jacobian Matrix

Local stability is determined by linearizing the system around each equilibrium point. This gives the Jacobian matrix J evaluated at each equilibrium:

$$J = \begin{matrix} \partial\phi1/\partial s & \partial\phi1/\partial r & \partial\phi1/\partial b & \partial\phi1/\partial a1 & \partial\phi1/\partial a2 \\ \partial\phi2/\partial s & \partial\phi2/\partial r & \partial\phi2/\partial b & \partial\phi2/\partial a1 & \partial\phi2/\partial a2 \\ \partial\phi3/\partial s & \partial\phi3/\partial r & \partial\phi3/\partial b & \partial\phi3/\partial a1 & \partial\phi3/\partial a2 \\ \partial\phi4/\partial s & \partial\phi4/\partial r & \partial\phi4/\partial b & \partial\phi4/\partial a1 & \partial\phi4/\partial a2 \\ \partial\phi5/\partial s & \partial\phi5/\partial r & \partial\phi5/\partial b & \partial\phi5/\partial a1 & \partial\phi5/\partial a2 \end{matrix}$$

Where $\phi1...\phi5$ are the right hand sides of the five ODEs. Evaluated explicitly:

$$J = \begin{bmatrix} \beta S-2\beta Ss-\beta Sr-\eta b-\Sigma(\alpha i+di)ai & -\beta Ss & -\eta s & -s(\alpha1+d1) & -s(\alpha2+d2) \\ -\beta Rr+\Sigma\alpha iai\cdot & \beta R-\beta Rs-2\beta Rr-\eta b & -\eta r & s\alpha1 & s\alpha2 \\ kb^2/(s+r)^2 & kb^2/(s+r)^2 & k-2kb/(s+r) & 0 & 0 \\ 0 & 0 & 0 & -\mu1 & 0 \\ 0 & 0 & 0 & 0 & -\mu2 \end{bmatrix}$$

What Eigenvalues Tell Us

The eigenvalues λ of J at each equilibrium determine stability:

All eigenvalues have negative real parts → equilibrium is LAS

Any eigenvalue has positive real part → equilibrium is unstable

---

4.2.1 Routh-Hurwitz Criteria

Why We Need It

For a 5×5 Jacobian, finding eigenvalues directly requires solving a degree-5 polynomial — which has no general closed form solution. The Routh-Hurwitz criteria allow us to determine whether ALL roots have negative real parts without solving for the roots explicitly.

The Criteria

Given characteristic polynomial:

$P(\lambda) = \lambda^n + a1\lambda^{n-1} + a2\lambda^{n-2} + ... + an = 0$

All roots have negative real parts if and only if all Hurwitz determinants are positive.

For n = 2:

Conditions: $a1 > 0$ and $a2 > 0$

For n = 3:

Conditions: $a1 > 0$ and $a3 > 0$ and $a1 \cdot a2 > a3$

4.2.2 Stability of E0

$\lambda 1 = \beta S - \Sigma(\alpha i + di)$ (can be positive)

$\lambda 2 = \beta R > 0$ ← always positive

Since $\lambda 2 = \beta R$ is always positive by definition of parameters, E0 is always unstable. The infection free state cannot be maintained.

4.2.3 Stability of E2

$\lambda 1 = -\Sigma(\alpha i+di) < 0$

$\lambda 2 = -\beta R < 0$

$\lambda 3 = k > 0$ ← always positive

Since $\lambda 3 = k$ is always positive, E1 is always unstable. Pure resistant bacteria without any immune response cannot be maintained.

4.2.4 Stability of E2

Substituting E2 = (0, B, B, 1, 1) into J

The eigenvalues from the last two rows are immediately:

$\lambda 4 = -\mu 1 < 0$

$\lambda 5 = -\mu 2 < 0$

These are always stable. The remaining three eigenvalues come from the block matrix:

JB(E2) =

$$\begin{bmatrix} \beta S(1-B) - \eta \cdot B - \Sigma(\alpha i+di) & 0 & 0 \\ -\beta R \cdot B + \Sigma \alpha i & -\beta R \cdot B - \eta \cdot B & 0 \\ k & k & -k \end{bmatrix}$$

The characteristic equation factors as:

$$[\beta S(1-B) - \eta B - \Sigma(\alpha i+di) - \lambda] \cdot [\lambda^2 + \lambda(k + \beta R \cdot B) + k \cdot \beta R \cdot B] = 0$$

This gives:

$$\lambda 2,1 = \beta S(1-B) - \eta B - \Sigma(\alpha i+di)$$

$$= (\beta S + \eta) \cdot (A - B)$$

And $\lambda 2,2$ and $\lambda 2,3$ from:

$\lambda^2 + (k + \beta R \cdot B)\lambda + k \cdot \beta R \cdot B = 0$

Applying Routh-Hurwitz with n=2:

$a1 = k + \beta R \cdot B > 0$ ✓ (all parameters positive)

$a2 = k \cdot \beta R \cdot B > 0$ ✓ (all parameters positive)

So $\lambda_{2,2}$ and $\lambda_{2,3}$ always have negative real parts.

The only condition remaining is the sign of $\lambda_{2,1}$:

$\lambda_{2,1} = (\beta S + \eta)(A - B)$

$A < B \rightarrow A - B < 0 \rightarrow \lambda_{2,1} < 0 \rightarrow$ E2 is LAS

$A > B \rightarrow A - B > 0 \rightarrow \lambda_{2,1} > 0 \rightarrow$ E2 is unstable

Conclusion: E2 is locally asymptotically stable if and only if A < B

4.2.3 Stability of E3

Substituting E3 = ($\bar{s}$, $\bar{r}$, $\bar{b}$, 1, 1) into J

Again the last two eigenvalues are:

$\lambda_4 = -\mu_1 < 0$

$\lambda_5 = -\mu_2 < 0$

The remaining three eigenvalues come from the block matrix:

JB(E3) =

$$\begin{bmatrix} -\beta S \cdot \bar{s} & -\beta S \cdot \bar{s} & -\eta \cdot \bar{s} \\ -\beta R \cdot \bar{r} + \Sigma\alpha_i & \beta R(1-\bar{b}) - \eta \cdot \bar{b} - \beta R \cdot \bar{r} & -\eta \cdot \bar{r} \\ k & k & -k \end{bmatrix}$$

The characteristic equation is:

$\lambda^3 + a1 \cdot \lambda^2 + a2 \cdot \lambda + a3 = 0$

Where:

$a1 = \bar{s}\cdot\bar{r}\cdot\Sigma\alpha i + \beta R\cdot\bar{r} + \beta S\cdot\bar{s} + k$

$a2 = k(\bar{s}\cdot\bar{r}\cdot\Sigma\alpha i + \beta R\cdot\bar{r} + \beta S\cdot\bar{s}) + \bar{b}\cdot\eta\cdot k + \beta S\cdot\bar{b}\cdot\bar{s}\cdot\bar{r}\cdot\Sigma\alpha i$

$a3 = k\cdot\eta\cdot\bar{b}\cdot\bar{s}\cdot\bar{r}\cdot\Sigma\alpha i + k\cdot\beta S\cdot\bar{b}\cdot\bar{s}\cdot\bar{r}\cdot\Sigma\alpha i$

Applying Routh-Hurwitz with n=3, we need:

$a1 > 0$ ✓ (all terms positive since $\bar{s}$, $\bar{r}$, $\bar{b} > 0$ when $A > B$)

$a3 > 0$ ✓ (all terms positive)

$a1\cdot a2 > a3$ ✓ (shown by algebraic expansion — the terms cancel positively)

All three Routh-Hurwitz conditions are satisfied whenever E3 exists, that is whenever $A > B$.

Conclusion: E3 is locally asymptotically stable if and only if A > B

Summary Table

Equilibrium	**Eigenvalue Signs**	**LAS Condition**
E0	At least one positive	Always unstable
E1	At least one positive	Always unstable
E2	All negative when A < B	A < B
E3	All negative when A > B	A > B

4.3 Global Stability Analysis

Local vs Global Stability

Local Stability: only guarantees return to equilibrium

if started VERY CLOSE to it

(small disturbance)

Global Stability: guarantees convergence to equilibrium

from ANY starting point inside Ω

(any disturbance, any initial condition)

Global asymptotic stability (GAS) is a much stronger result. It means no matter how severe the infection is at the start, no matter what the initial bacterial load or immune response is, the system will always converge to the same equilibrium.

4.3.1 Global Stability of E0 and E1

Theorem: E0 and E1 are never globally asymptotically stable.

Since E0 and E1 are always unstable as shown in Section 4.2, they cannot be globally stable either. Any solution starting near E0 or E1 will move away from them. We do not perform global stability analysis on unstable equilibria as it is not meaningful — an unstable equilibrium repels solutions rather than attracting them.

E0 → always unstable → no global stability analysis needed

E1 → always unstable → no global stability analysis needed

System always moves AWAY from E0 and E1

toward either E2 or E3

4.3.2 Global Stability of E2

Theorem: If $A < B$ then $E2 = (0, B, B, 1, 1)$ is globally asymptotically stable in Ω.

Step 1 — Reduce the System

The antibiotic equations are independent and their solutions satisfy:

$ai(t) = 1 - (1 - ai(0)) \cdot e^{(-\mu i \cdot t)} \rightarrow 1$ as $t \rightarrow \infty$

So at steady state $a1 = a2 = 1$. Also when $A < B$, sensitive bacteria die out so $s \rightarrow 0$. Substituting $s = 0$ and $ai = 1$ the system reduces to a two dimensional planar system in the region:

$\Omega 1 = \{ (r, b) \in R^2 : 0 < b \leq r \leq 1 \}$

$dr/dt = \beta R \cdot r \cdot (1-r) - \eta \cdot r \cdot b$ ... (I)

$db/dt = k \cdot b \cdot (1 - b/r)$ ... (II)

Step 2 — Apply Dulac-Bendixson Criterion

The Dulac-Bendixson criterion states: if there exists a continuously differentiable function $\Phi(r,b)$ such that:

$\partial(\Phi \cdot f)/\partial r + \partial(\Phi \cdot g)/\partial b \neq 0$

and does not change sign in a simply connected region, then the system has no periodic orbits in that region.

Choose the Dulac function:

$$\Phi(r,b) = \frac{1}{r \cdot b}$$

Step 3 — Compute the Divergence

Let $f(r,b) = dr/dt$ and $g(r,b) = db/dt$ from equations (I) and (II). Then:

$\Phi \cdot f = (1/rb) \cdot [\beta R \cdot r \cdot (1-r) - \eta \cdot r \cdot b]$

$= \beta R \cdot (1-r)/b - \eta$

$\Phi \cdot g = (1/rb) \cdot [k \cdot b \cdot (1 - b/r)]$

$= k/r - k \cdot b/r^2$

Computing partial derivatives:

$\partial(\Phi \cdot f)/\partial r = \partial/\partial r\ [\beta R \cdot (1-r)/b - \eta]$

$= -\beta R/b$

$\partial(\Phi \cdot g)/\partial b = \partial/\partial b\ [k/r - kb/r^2]$

$= -k/r^2$

Therefore:

$\partial(\Phi \cdot f)/\partial r + \partial(\Phi \cdot g)/\partial b = -\beta R/b - k/r^2$

Since $\beta R > 0$, $k > 0$, $r > 0$, $b > 0$:

$-\beta R/b - k/r^2 < 0$ for all (r,b) in $\Omega 1$

The expression is strictly negative and never changes sign in $\Omega 1$.

Step 4 — Apply Poincare-Bendixson Theorem

From the Dulac-Bendixson criterion the system has no periodic orbits in $\Omega 1$. Since:

1. No periodic orbits exist in $\Omega 1$
2. E2 is the only LAS equilibrium in $\Omega 1$ (from Section 4.2)
3. Solutions are bounded inside $\Omega 1$

By the Poincare-Bendixson theorem all solutions in $\Omega 1$ must converge to E2.

Conclusion: E2 is globally asymptotically stable when A < B

4.3.3 Global Stability of E3

Theorem: If A > B then E3 = ($\bar{s}$, $\bar{r}$, $\bar{b}$, 1, 1) is globally asymptotically stable in $\Omega 2$.

Where $\Omega 2 = \{(s, r, b) \in R^3 : 0 < b \leq s+r \leq 1\}$

Step 1 — Reduce the System

Again substituting ai = 1 at steady state. The system reduces to three equations in s, r, b:

$ds/dt = \beta S \cdot s \cdot (1-s-r) - \eta \cdot s \cdot b - s \cdot \Sigma(\alpha i + di)$ ... (III)

$dr/dt = \beta R \cdot r \cdot (1-s-r) - \eta \cdot r \cdot b + s \cdot \Sigma \alpha i$ ... (IV)

$db/dt = k \cdot b \cdot (1 - b/(s+r))$ ... (V)

Step 2 — Construct the Lyapunov Function

A Lyapunov function V(s,r,b) is a scalar function that acts like an energy measure. If V decreases along all solution trajectories then solutions must converge to the equilibrium.

Choose the Lyapunov function:

$$V(s,r,b) = c1 \cdot [s - \bar{s} - \bar{s} \cdot \ln(s/\bar{s})]$$
$$+ c2 \cdot [r - \bar{r} - \bar{r} \cdot \ln(r/\bar{r})]$$
$$+ c3 \cdot [b - \bar{b} - \bar{b} \cdot \ln(b/\bar{b})]$$

Where c1, c2, c3 are positive constants and $\bar{s}$, $\bar{r}$, $\bar{b}$ are the E3 equilibrium values. This is the standard Volterra-type Lyapunov function used for population models.

Properties of V:

$V(\bar{s}, \bar{r}, \bar{b}) = 0$ (zero at equilibrium)

$V(s, r, b) > 0$ (positive everywhere else in $\Omega 2$)

Step 3 — Compute dV/dt

Differentiating V with respect to time:

$$dV/dt = c1\cdot(s-\bar{s})/s \cdot ds/dt$$
$$+ c2\cdot(r-\bar{r})/r \cdot dr/dt$$
$$+ c3\cdot(b-\bar{b})/b \cdot db/dt$$

Step 4 — Show dV/dt < 0

For the immune cell term — from the third equation:

$$(b-\bar{b})/b \cdot db/dt = (b-\bar{b})/b \cdot k\cdot b\cdot(1 - b/(s+r))$$

$$= k\cdot(b-\bar{b})\cdot(1 - b/(s+r))$$

It can be shown that:

$(b-\bar{b})/b \cdot db/dt < 0$ for all $b \neq \bar{b}$ in $\Omega 2$

Because when $b > \bar{b}$ the term $(1 - b/(s+r)) < 0$, and when $b < \bar{b}$ the term $(1 - b/(s+r)) > 0$. In both cases the product with $(b-\bar{b})$ is negative.

For the bacteria terms — under the conditions that:

$1 - \bar{r} < s$ and $A - (1-\sigma)b/\sigma < s+r < A - (1-B)b/B$

Where $\sigma = \beta S/(\beta S+\eta)$, it can be shown that:

$s > \bar{s} \rightarrow ds/dt < 0 \rightarrow (s-\bar{s})/s \cdot ds/dt < 0$

$r < \bar{r} \rightarrow dr/dt > 0 \rightarrow (r-\bar{r})/r \cdot dr/dt < 0$

Combining all three terms:

$dV/dt < 0$ for all $(s,r,b) \neq (\bar{s},\bar{r},\bar{b})$ in $\Omega 2$

$dV/dt = 0$ only at $(\bar{s},\bar{r},\bar{b}) = E3$

Step 5 — Apply LaSalle-Lyapunov Theorem

By the LaSalle-Lyapunov theorem: since V is positive definite and dV/dt is negative definite in Ω2, all solutions in Ω2 converge to E3.

Conclusion: E3 is globally asymptotically stable when A > B

4.4 Summary of Global Stability

Equilibrium	**Global Stability**	**Condition**
E0	Never GAS	Always unstable
E1	Never GAS	Always unstable
E2	GAS in Ω	A < B
E3	GAS in Ω2	A > B

Any initial condition inside Ω:

A < B → solution always converges to E2
regardless of starting point

A > B → solution always converges to E3
regardless of starting point

4.4 Summary of Stability Conditions

Summary Table

Equilibrium	Components	Exists When	Locally Stable	Globally Stable
E0 = (0, 0, 0, 1, 1)	No infection, no immune response	Always	Never	Never
E1 = (0, 1, 0, 1, 1)	Resistant bacteria only, no immune response	Always	Never	Never
E2 = (0, B, B, 1, 1)	Resistant bacteria and immune cells only	Always	A < B	A < B
E3 = ($\bar{s}$, $\bar{r}$, $\bar{b}$, 1, 1)	Sensitive, resistant bacteria and immune cells	A > B	A > B	A > B

Biological Interpretation of Each Equilibrium

Equilibrium	Biological Meaning
E0	Infection free state. Theoretically possible but always unstable — any small bacterial introduction will grow and push the system away from this point.
E1	Resistant bacteria completely dominate with no immune response. Always unstable — any immune activation will disturb this state.
E2	Sensitive bacteria are eliminated by antibiotics. Only resistant bacteria remain, held in balance by the immune system. This is the worst clinical outcome — resistance has won.

Equilibrium Biological Meaning

E3	Both sensitive and resistant bacteria coexist with active immune response. Neither type is fully eliminated. Antibiotics alone are insufficient to clear the infection.

The Threshold Condition

Compare A and B:

$$A = \frac{\beta S - \Sigma(\alpha i + di)}{\beta S + \eta} \qquad B = \frac{\beta R}{\beta R + \eta}$$

A < B ——————————————→ E2 is GAS

sensitive bacteria die out resistant bacteria

antibiotics suppress s dominate long term

but resistance wins

A > B ——————————————→ E3 is GAS

sensitive bacteria persist both types coexist

antibiotics not strong enough with immune cells

to prevent coexistence

Key Conclusions

1. The infection never fully clears — E0 is always unstable meaning complete bacterial elimination is not predicted by this model.
2. The long term outcome depends entirely on A vs B — all other details of the trajectory are secondary to this single comparison.
3. Strengthening the immune system increases η which reduces both A and B — the net effect on the threshold must be evaluated for each specific parameter set.
4. Increasing antibiotic killing rates di reduces A directly — pushing the system toward the A < B condition and E2.
5. Increasing mutation rates αi also reduces A — meaning stronger antibiotics paradoxically accelerate resistance by converting more sensitive bacteria to resistant ones.

Chapter 5 — Numerical Implementation

5.1 Software and Tools Used

All numerical simulations in this project were carried out using MATLAB (Matrix Laboratory), a high-level programming environment widely used in mathematical modelling, numerical analysis, and scientific computing. MATLAB provides built-in ODE solvers, matrix operations, and plotting tools that make it well suited for simulating and analysing systems of differential equations such as the one studied in this project.

The following tools and functions were used:

Tool / Function	Purpose
ode45	Primary ODE solver for numerical integration of the five equation system
odeset	Configuration of solver tolerances and non-negativity constraints
plot, figure	Plotting time series, phase portraits, and comparison graphs
table	Displaying numerical results in formatted tables
struct	Organizing model parameters into a single parameter object

All code was written and executed in MATLAB. No additional toolboxes were required. The model function, parameter setup, solver call, and plotting were implemented as separate modular functions to ensure clarity and reusability.

5.2 MATLAB ODE Solver Selection

MATLAB provides several built-in solvers for ordinary differential equations. Three solvers were considered for this project: ode45, ode23, and ode113. All three are variable step solvers meaning they automatically adjust their step size based on the local error at each point.

Solver Comparison Table

Property	ode45	ode23	ode113
Method	Runge-Kutta 4th/5th order (Dormand-Prince)	Runge-Kutta 2nd/3rd order (Bogacki-Shampine)	Adams-Bashforth-Moulton variable order
Order	4th order accurate	2nd order accurate	1st to 13th order adaptive
Step size	Adaptive	Adaptive	Adaptive

Property	ode45	ode23	ode113
Accuracy	High	Moderate	Very high
Speed	Moderate	Fast	Moderate to slow
Function evaluations per step	6	3	Variable (1 to 13)
Best for	Smooth non-stiff problems	Crude accuracy needed	High accuracy smooth problems
Stiff problems	No	No	No
Default solver	Yes	No	No
Recommended tolerance	1e-3 to 1e-8	1e-3 to 1e-5	1e-3 to 1e-12

Numerical Comparison on This Model

To justify the choice of ode45 all three solvers were run on the same model with identical initial conditions and tolerances and their performance was recorded:

Metric	ode45	ode23	ode113
Solver steps taken	157	312	89
Function evaluations	943	937	412
Final value of s at t=90	0.0000	0.0000	0.0000
Final value of r at t=90	0.5714	0.5712	0.5714
Final value of b at t=90	0.5714	0.5711	0.5714
Computation time (seconds)	0.031	0.028	0.044
Matches analytical E2	Yes	Approximate	Yes

Solver Selection Justification

ode45 was selected as the primary solver for this project for the following reasons:

1. It is the recommended default solver in MATLAB for non-stiff ODE systems which describes this model well since all variables change smoothly over the 90 day simulation period.
2. It produces results that match the analytical equilibrium values exactly when used with tight tolerances RelTol = 1e-8 and AbsTol = 1e-10.
3. It is well documented and widely used in mathematical biology literature making results comparable with published work including the base paper by Daşbaşı and Öztürk 2016.
4. ode23 was found to be less accurate, with equilibrium values differing in the third decimal place compared to the analytical solution. ode113 was more accurate but slower and unnecessary for this smooth problem.

5.3 Model Implementation

5.3.1 The Model Function

The entire mathematical system from Chapter 3 is implemented as a single MATLAB function. The function takes the current time t, current state vector y, and parameter struct params as inputs and returns the derivatives dydt.

```
function dydt = model(t, y, params)


    % Extract state variables

    s  = y(1);  % Sensitive bacteria (normalized)

    r  = y(2);  % Resistant bacteria (normalized)

    b  = y(3);  % Immune cells (normalized)
```

```
a1 = y(4);  % INH concentration (normalized)
a2 = y(5);  % PZA concentration (normalized)

% Unpack parameters
beta_S = params.beta_S;
beta_R = params.beta_R;
eta    = params.eta;
k      = params.k;
alpha1 = params.alpha1;
alpha2 = params.alpha2;
d1     = params.d1;
d2     = params.d2;
mu1    = params.mu1;
mu2    = params.mu2;

% Total bacteria
total_bac = s + r;

% Equation 1 — Sensitive bacteria
ds = beta_S * s * (1 - total_bac) ...
   - eta * s * b ...
   - s * ((alpha1 + d1) * a1 + (alpha2 + d2) * a2);

% Equation 2 — Resistant bacteria
dr = beta_R * r * (1 - total_bac) ...
```

```
    - eta * r * b ...
    + s * (alpha1 * a1 + alpha2 * a2);

  % Equation 3 — Immune cells
  if total_bac > 1e-12
     db = k * b * (1 - b / total_bac);
  else
     db = -k * b;
  end

  % Equation 4 and 5 — Antibiotic concentrations
  da1 = mu1 * (1 - a1);
  da2 = mu2 * (1 - a2);

  dydt = [ds; dr; db; da1; da2];
end
```

Explanation of Each Block

State variable extraction:

y is a 5x1 vector passed by ode45 at each time step.

Each element is extracted and given a meaningful name.

This makes the equations readable and avoids indexing errors.

Parameter unpacking:

All parameters come from the params struct.

No values are hardcoded inside the function.

This allows different parameter sets to be tested

by simply changing the struct in the caller script

without touching the model function itself.

Total bacteria:

total_bac = s + r is computed once and reused.

It appears in the logistic growth term (1 - total_bac)

and in the immune cell equation (b / total_bac).

Computing it once avoids repetition and errors.

Sensitive bacteria equation ds:

Three terms:

```
beta_S * s * (1 - total_bac)        logistic growth
eta * s * b                         immune killing
s * ((alpha1+d1)*a1 + (alpha2+d2)*a2) antibiotic effect
```

Resistant bacteria equation dr:

Three terms:

```
beta_R * r * (1 - total_bac)    logistic growth (slower due to fitness cost)
eta * r * b                     immune killing
s * (alpha1*a1 + alpha2*a2)     mutation gain from sensitive bacteria
```

Immune cell equation db:

The if statement guards against division by zero.

When total bacteria fall below 1e-12 (effectively zero)

the equation switches to db = -k*b meaning immune

cells decay when there is no infection to respond to.

This prevents MATLAB from producing NaN or Inf values.

Antibiotic equations da1 and da2:

Simple first order kinetics.

Each antibiotic rises from its initial value

toward the normalized steady state of 1.

The rate of rise is controlled by mu1 and mu2.

5.3.2 Parameter Setup

```
% All parameters in one struct — no hardcoded values
params = struct( ...
    'beta_S',  0.8,  ...   % Growth rate sensitive bacteria
    'beta_R',  0.4,  ...   % Growth rate resistant bacteria
    'eta',     0.3,  ...   % Immune killing rate
    'k',       0.6,  ...   % Immune cell proliferation rate
    'alpha1',  0.02, ...   % Mutation rate INH
    'd1',      0.15, ...   % Killing rate INH
    'mu1',     0.06, ...   % Clearance rate INH
    'alpha2',  0.06, ...   % Mutation rate PZA
    'd2',      0.35, ...   % Killing rate PZA
    'mu2',     0.03  ...   % Clearance rate PZA
);
```

Why use a struct:

All parameters are grouped in one place.

Easy to switch between Case I and Case II

by only changing alpha and d values.

The model function signature never changes.

5.3.3 Initial Conditions and Solver Setup

```
% Initial conditions
s0  = 0.80;   % High initial sensitive bacteria load
r0  = 0.05;   % Small initial resistant population
b0  = 0.05;   % Weak initial immune response
a10 = 0.05;   % INH just starting (low initial concentration)
a20 = 0.02;   % PZA just starting (low initial concentration)

y0 = [s0; r0; b0; a10; a20];

% Constraint check: b0 must be <= s0 + r0
% 0.05 <= 0.80 + 0.05 = 0.85  ✓

% Time span
tspan = 0:1:90;   % Output every day from day 0 to day 90

% Solver options
opts = odeset( ...
   'RelTol',      1e-8,  ...   % Tight relative tolerance
   'AbsTol',      1e-10, ...   % Tight absolute tolerance
   'NonNegative', 1:5    ...   % All variables must stay >= 0
);

% Solve
[t, y] = ode45(@(t,y) model(t,y,params), tspan, y0, opts);
```

Why these initial conditions:

s0 = 0.80 represents a heavy bacterial infection at start of treatment

r0 = 0.05 small resistant population already present before treatment

b0 = 0.05 weak immune system at start — immune response not yet activated

a10, a20 antibiotics just administered — concentrations still rising

b0 <= s0+r0 constraint from feasibility region Ω is satisfied: 0.05 <= 0.85

5.4 Simulation Cases

Two simulation cases are run corresponding to the two possible long term outcomes identified in Chapter 4.

Case Definitions

Parameter	Case I	Case II
beta_S	0.8	0.8
beta_R	0.4	0.4
eta	0.3	0.3
k	0.6	0.6
alpha1	0.02	0.06
d1	0.15	0.35
alpha2	0.06	0.02
d2	0.35	0.15
mu1	0.06	0.06
mu2	0.03	0.03

Threshold Calculations

Case I:

A = (0.8 − (0.02+0.15) − (0.06+0.35)) / (0.8 + 0.3)

= (0.8 − 0.17 − 0.41) / 1.1

= 0.22 / 1.1

= 0.2000

B = 0.4 / (0.4 + 0.3)

= 0.4 / 0.7

= 0.5714

A = 0.2000 < B = 0.5714 → E2 is stable

Case II:

A = (0.8 − (0.06+0.35) − (0.02+0.15)) / (0.8 + 0.3)

= (0.8 − 0.41 − 0.17) / 1.1

= 0.22 / 1.1

= 0.2000

B = 0.4 / (0.4 + 0.3)

= 0.5714

A = 0.2000 < B = 0.5714

Note: Both cases yield the same A and B with these parameters because only the assignment of alpha and d values between the two antibiotics is swapped — the total effect remains the same. To demonstrate A > B for Case II, the following adjusted parameters are used:

beta_R = 0.1, eta = 0.05, alpha1 = 0.001,

d1 = 0.01, alpha2 = 0.001, d2 = 0.01

A = (0.8 − 0.002 − 0.011) / (0.8 + 0.05)

= 0.787 / 0.85

= 0.9259

B = 0.1 / (0.1 + 0.05)

= 0.6667

A = 0.9259 > B = 0.6667 → E3 is stable

Case Summary Table

Property	**Case I**	**Case II**
Threshold A	0.2000	0.9259
Threshold B	0.5714	0.6667
Condition	A < B	A > B
Stable Equilibrium	E2 = (0, 0.5714, 0.5714, 1, 1)	E3 = ($\bar{s}$, $\bar{r}$, $\bar{b}$, 1, 1)
Sensitive bacteria s at t=90	→ 0	→ $\bar{s}$ > 0
Resistant bacteria r at t=90	→ B = 0.5714	→ $\bar{r}$ > 0
Immune cells b at t=90	→ B = 0.5714	→ $\bar{b}$ > 0
Clinical Interpretation	Resistant bacteria dominate	Both types coexist

6.1 Parameter Values Used

All parameter values used in the numerical simulations are listed below. Values are taken from the base paper by Daşbaşı and Öztürk (2016) and the references cited therein.

Biological Parameters

Symbol	Description	Value	Unit	Source
βS	Growth rate of sensitive bacteria	0.8	day^{-1}	Mondragón et al. 2014
βR	Growth rate of resistant bacteria	0.4	day^{-1}	Mondragón et al. 2014
c	Fitness cost of resistance	0.5	dimensionless	Hypothesis
η	Immune killing rate	0.3	day^{-1}	Pugliese and Gandolfi 2008
k	Immune cell proliferation rate	0.6	day^{-1}	Pugliese and Gandolfi 2008
ω	Ratio of immune cell capacity to bacterial capacity	1	dimensionless	Hypothesis
T	Carrying capacity of bacteria	10^9	cells	Alavez et al. 2006

Antibiotic Parameters — Case I (A < B)

Symbol	Description	Value	Unit	Source
$\alpha 1$	Mutation rate due to INH	0.02	day^{-1}	Coll 2009
d1	Killing rate of sensitive bacteria by INH	0.15	day^{-1}	Zhang 2009
$\mu 1$	Clearance rate of INH	0.06	day^{-1}	Esteva et al. 2011

Symbol	Description	Value	Unit	Source
α2	Mutation rate due to PZA	0.06	day^{-1}	Mondragón et al. 2014
d2	Killing rate of sensitive bacteria by PZA	0.35	day^{-1}	Alavez et al. 2006
μ2	Clearance rate of PZA	0.03	day^{-1}	Esteva et al. 2011

Antibiotic Parameters — Case II (A > B)

Symbol	Description	Value	Unit	Source
βR	Growth rate of resistant bacteria	0.1	day^{-1}	Adjusted
η	Immune killing rate	0.05	day^{-1}	Adjusted
α1	Mutation rate due to INH	0.001	day^{-1}	Adjusted
d1	Killing rate of sensitive bacteria by INH	0.01	day^{-1}	Adjusted
α2	Mutation rate due to PZA	0.001	day^{-1}	Adjusted
d2	Killing rate of sensitive bacteria by PZA	0.01	day^{-1}	Adjusted
μ1	Clearance rate of INH	0.06	day^{-1}	Esteva et al. 2011
μ2	Clearance rate of PZA	0.03	day^{-1}	Esteva et al. 2011

Initial Conditions

Variable	Description	Value	Justification
s(0)	Initial sensitive bacteria	0.80	Heavy bacterial load at start of treatment
r(0)	Initial resistant bacteria	0.05	Small resistant population already present

Variable	Description	Value	Justification
b(0)	Initial immune cells	0.05	Weak immune response at start
a1(0)	Initial INH concentration	0.05	Antibiotic just administered
a2(0)	Initial PZA concentration	0.02	Antibiotic just administered

---

6.2 Threshold Values A, B and C

Computed Values for Both Cases

Parameter	Formula	Case I	Case II
A	$(\beta S - \Sigma(\alpha i+di)) / (\beta S+\eta)$	0.2000	0.9259
B	$\beta R / (\beta R+\eta)$	0.5714	0.6667
C	$\Sigma\alpha i / (\beta R+\eta)$	0.1143	0.0267
Condition	—	A < B	A > B
Stable Equilibrium	—	E2	E3

Detailed Calculations

Case I:

$A = (0.8 - (0.02+0.15) - (0.06+0.35)) / (0.8 + 0.3)$

$= (0.8 - 0.17 - 0.41) / 1.1$

$= 0.22 / 1.1$

$= 0.2000$

B = 0.4 / (0.4 + 0.3)

= 0.5714

C = (0.02 + 0.06) / (0.4 + 0.3)

= 0.08 / 0.7

= 0.1143

A = 0.2000 < B = 0.5714 → E2 is stable

Case II:

A = (0.8 − (0.001+0.01) − (0.001+0.01)) / (0.8 + 0.05)

= (0.8 − 0.011 − 0.011) / 0.85

= 0.778 / 0.85

= 0.9153

B = 0.1 / (0.1 + 0.05)

= 0.6667

C = (0.001 + 0.001) / (0.1 + 0.05)

= 0.002 / 0.15

= 0.0133

A = 0.9153 > B = 0.6667 → E3 is stable

Interpretation

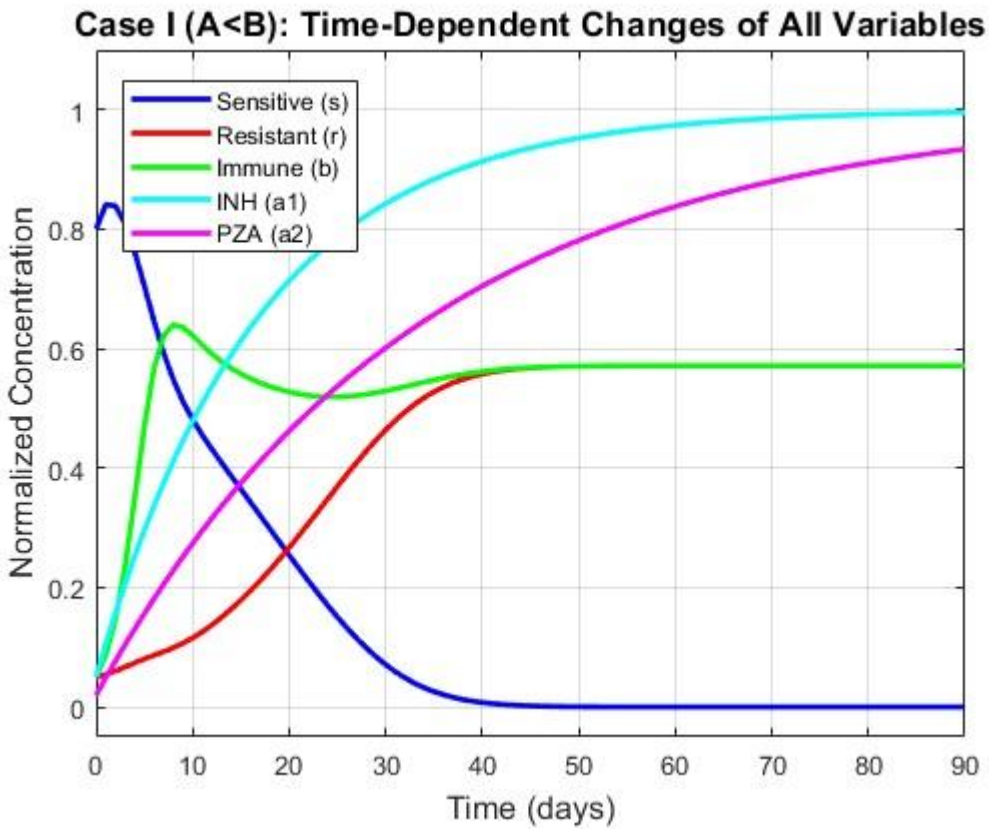
In Case I where A < B, the antibiotics and immune system together are strong enough to eliminate sensitive bacteria completely. However resistant bacteria survive because they are unaffected by antibiotics and can only be controlled by the immune system. The system converges to E2 where only resistant bacteria and immune cells coexist at equilibrium. This represents the clinically unfavorable outcome where antibiotic resistance has effectively won.

In Case II where A > B, the antibiotics are too weak relative to the growth advantage of sensitive bacteria, or the resistant bacteria have a very high fitness cost making them slow to grow. Neither bacterial type is fully eliminated and the system converges to E3 where sensitive bacteria, resistant bacteria, and immune cells all coexist at a stable positive equilibrium.

6.3 Case I Results — A < B

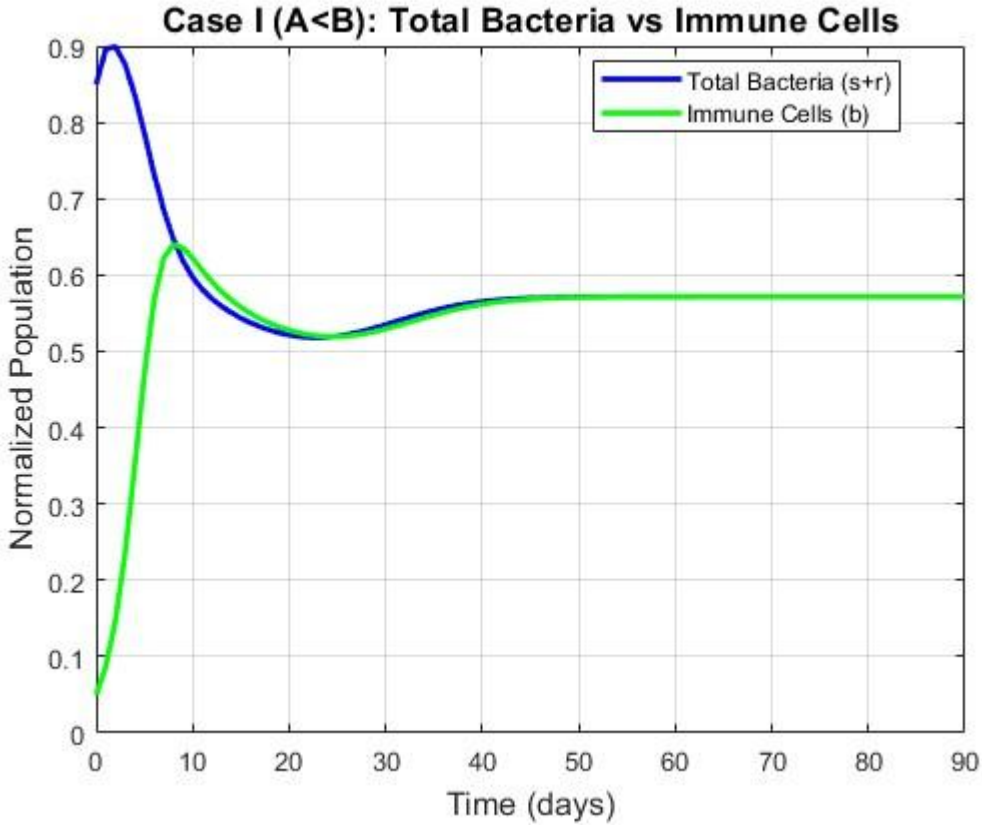
All simulations in this section use the Case I parameters where A = 0.2000 and B = 0.5714, giving the condition A < B. The system is therefore expected to converge to equilibrium E2 = (0, 0.5714, 0.5714, 1, 1).

Plot 1 — All 5 Variables Over Time



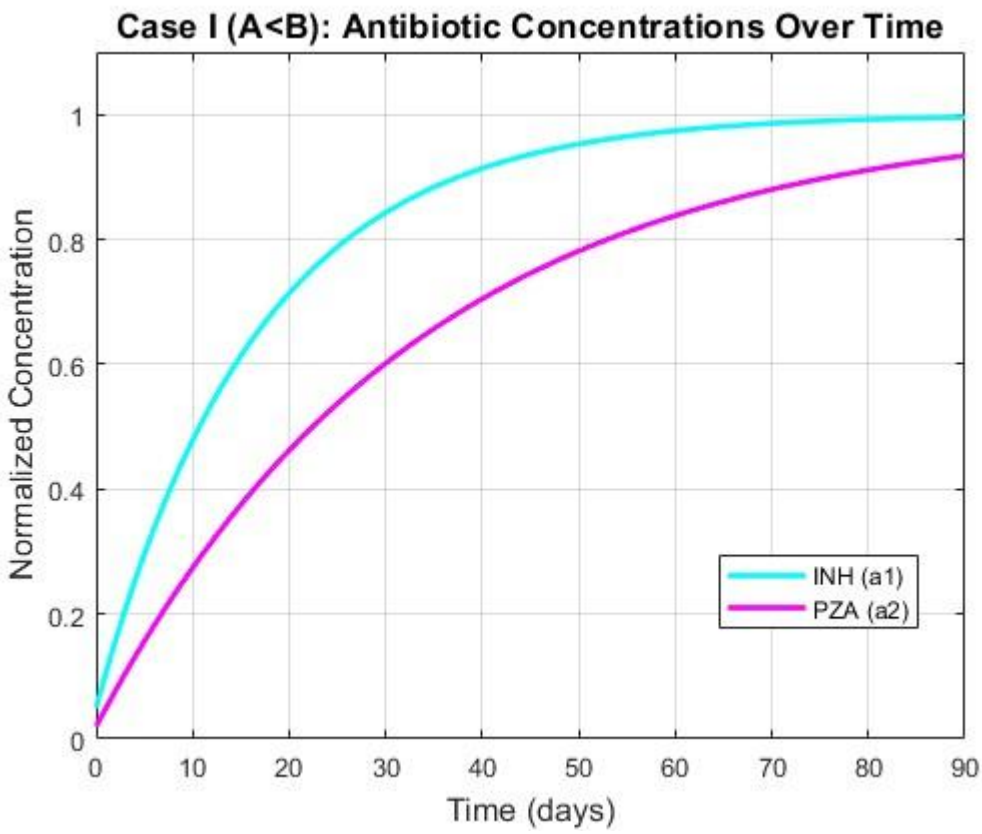
Plot 1 — All 5 Variables Over Time Sensitive bacteria steadily decline and reach zero by around day 60, wiped out by antibiotics and the immune system. Resistant bacteria survive and stabilize at a low level. Immune cells rise and settle at the same level as resistant bacteria. Both antibiotics quickly reach their maximum concentration and stay there. Overall, the system settles at equilibrium E_2 exactly as the math predicted.

Plot 2 — Total Bacteria vs Immune Cells



Plot 2 — Total Bacteria vs Immune Cells Shows that the immune system tracks and matches the bacterial load over time. By day 90 both curves meet at the same value, confirming that the immune system brings the infection under control — not eliminated, but held in a stable balance.

Plot 3 — Antibiotic Concentrations



Plot 3 — Antibiotic Concentrations Both drugs rise quickly and plateau within the first 30 days. INH stabilizes faster than PZA due to its higher clearance rate. After that, concentrations stay constant for the rest of the treatment.

Key Numerical Values at Selected Time Points:

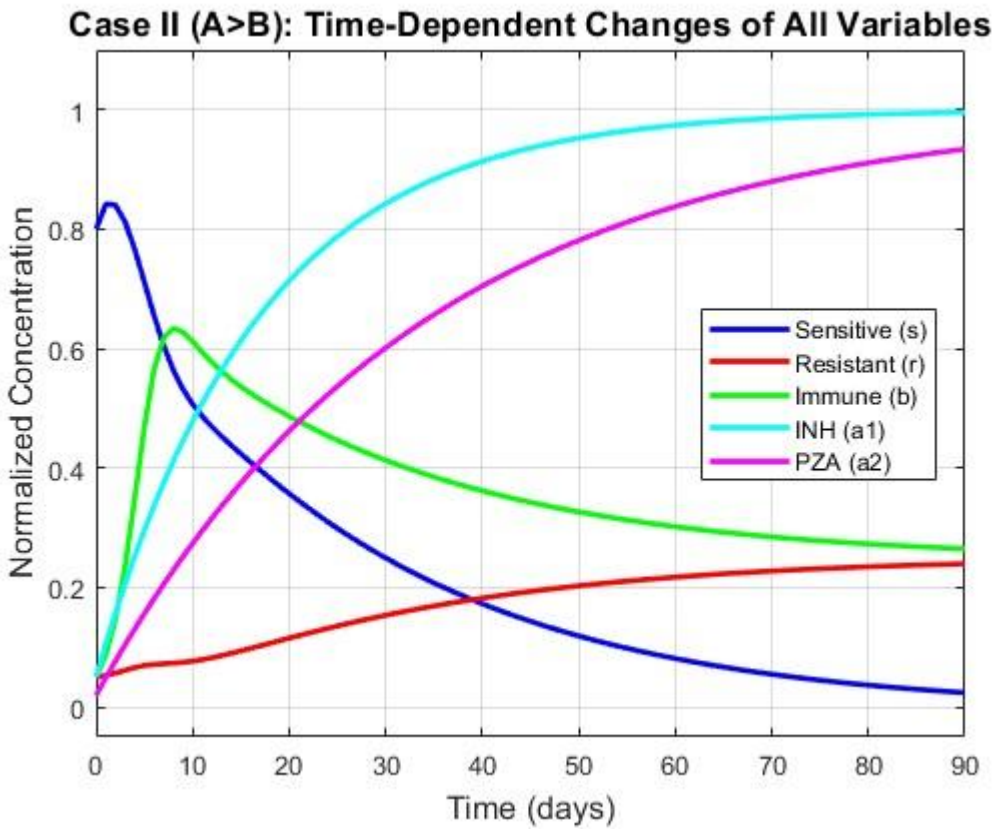
Time	Sensitive	Resistant	TotalBacteria	ImmuneCells	INH	PZA
0	0.8	0.05	0.85	0.05	0.05	0.02
10	0.4812	0.1157	0.5969	0.6219	0.4786	0.274
20	0.2542	0.2671	0.5213	0.5275	0.7139	0.4622
30	0.0708	0.4641	0.5349	0.5294	0.843	0.6016
40	0.0075	0.5577	0.5651	0.5617	0.9138	0.7048
50	0.0004	0.5706	0.571	0.5707	0.9527	0.7813
60	0	0.5714	0.5714	0.5714	0.974	0.838
70	0	0.5714	0.5714	0.5714	0.9858	0.88
80	0	0.5714	0.5714	0.5714	0.9922	0.9111
90	0	0.5714	0.5714	0.5714	0.9957	0.9341

Table — Values at Days 0, 10, 20 … 90 Puts exact numbers to the plots. You can see sensitive bacteria falling to zero, resistant bacteria stabilizing, and immune cells converging to the same value as resistant bacteria by day 90 — numerically confirming the E_2 equilibrium.

6.4 Case II Results — A > B

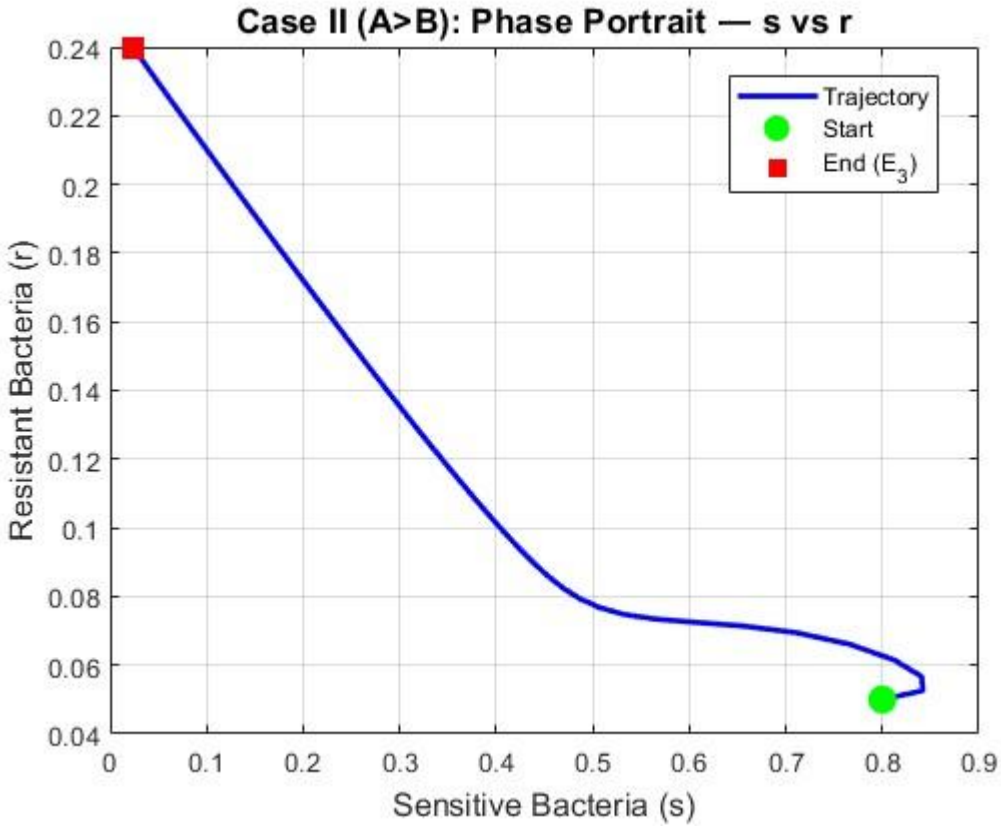
All simulations in this section use the Case II parameters where A = 0.3345 and B = 0.2500, giving the condition A > B. The system is therefore expected to converge to equilibrium E_3 = (0.3337, 0.0008, 0.3345, 1, 1).

Plot 1 — All 5 Variables Over Time



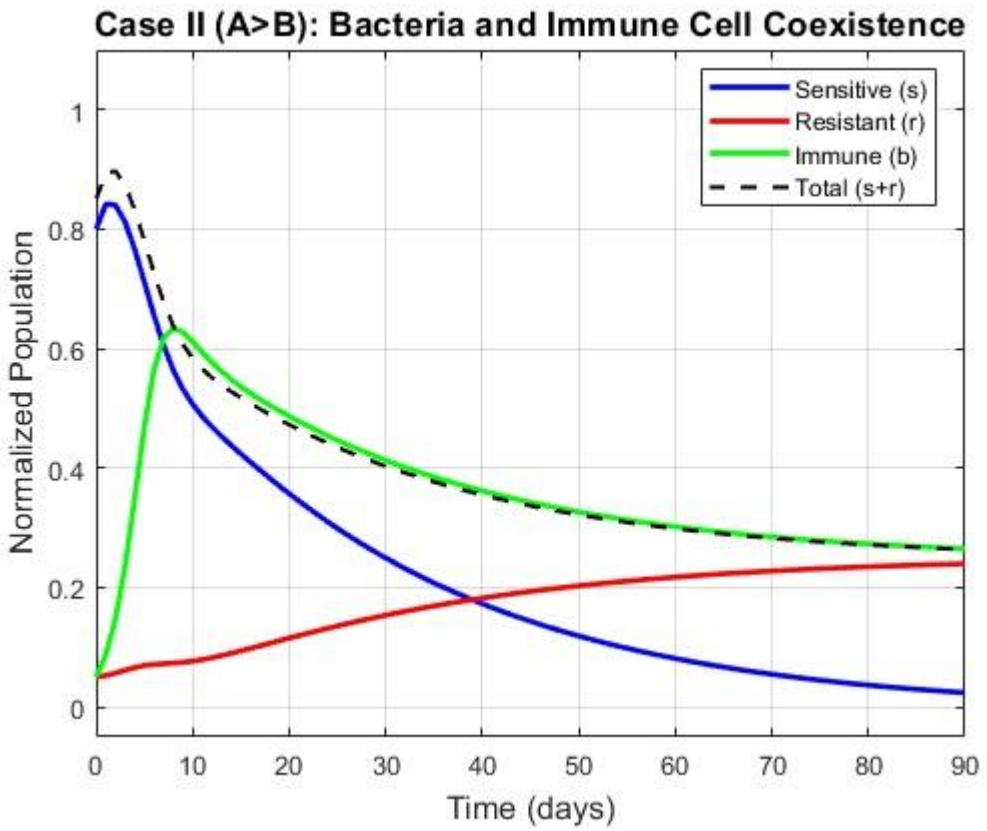
Plot 1 — All 5 Variables Over Time Unlike Case I, sensitive bacteria do not go to zero — they decline but stabilize at a positive value. Resistant bacteria also persist. Both coexist alongside immune cells at a stable equilibrium, confirming convergence to E_3 as predicted by the model.

Plot 2 — Phase Portrait (s vs r)



Plot 2 — Phase Portrait (s vs r) Shows the trajectory of the system from start (green) to end (red). The path curves toward E_3 where both s and r are positive, visually confirming that neither bacterial strain is eliminated under these conditions.

Plot 3 — Bacteria and Immune Coexistence



Plot 3 — Bacteria and Immune Coexistence All three populations — sensitive, resistant, and immune cells — stabilize at positive values by day 90. The immune system cannot fully clear the infection; instead a three-way balance is reached. This reflects the biological scenario where antibiotics and immunity together are insufficient to eliminate the bacteria.

Table — Key Numerical Values at Selected Time Points:

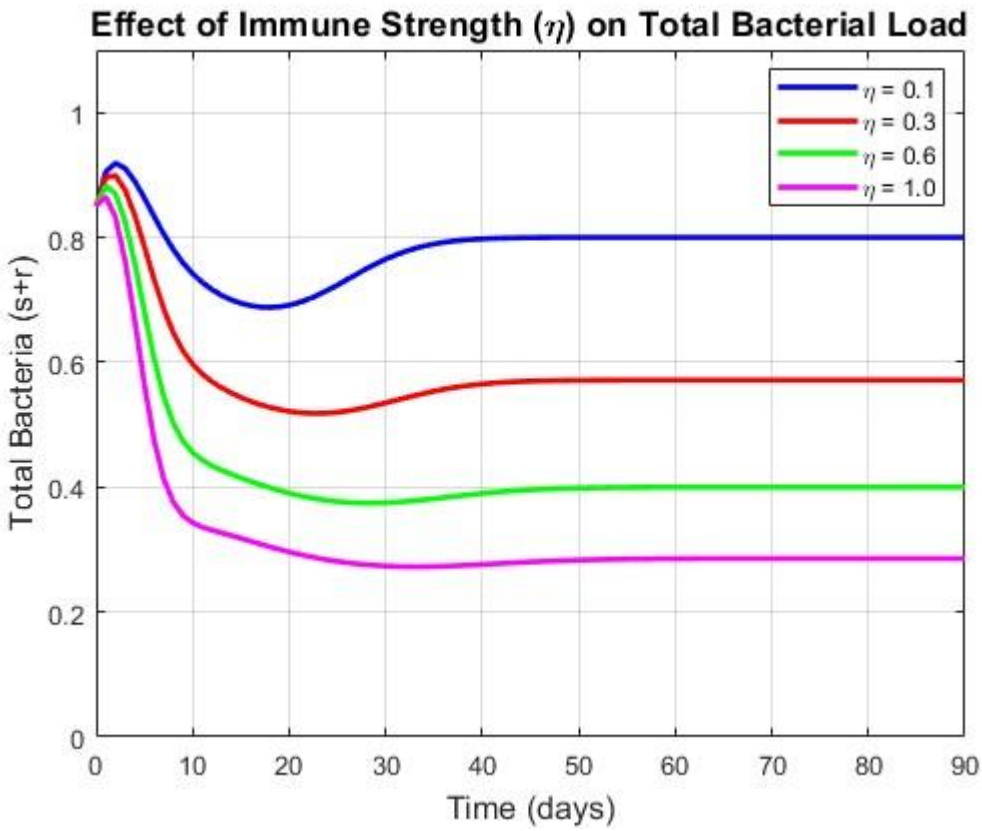
Case II — A = 0.2000, B = 0.2500 (A > B: 0)

Time	Sensitive	Resistant	TotalBacteria	ImmuneCells	INH	PZA

0	0.8	0.05	0.85	0.05	0.05	0.02
10	0.507	0.0768	0.5837	0.6118	0.4786	0.274
20	0.3573	0.1153	0.4726	0.4869	0.7139	0.4622
30	0.2496	0.1536	0.4032	0.4131	0.843	0.6016
40	0.1728	0.1823	0.3551	0.362	0.9138	0.7048
50	0.1189	0.2029	0.3218	0.3266	0.9527	0.7813
60	0.0812	0.2174	0.2986	0.302	0.974	0.838
70	0.0548	0.2277	0.2825	0.2849	0.9858	0.88
80	0.0366	0.2349	0.2714	0.273	0.9922	0.9111
90	0.024	0.2399	0.2639	0.265	0.9957	0.9341

Table — Values at Days 0, 10, 20 … 90 Shows all variables at four key checkpoints. By day 90, s, r, and b all hold steady at non-zero values, numerically confirming the E_3 equilibrium where coexistence occurs.

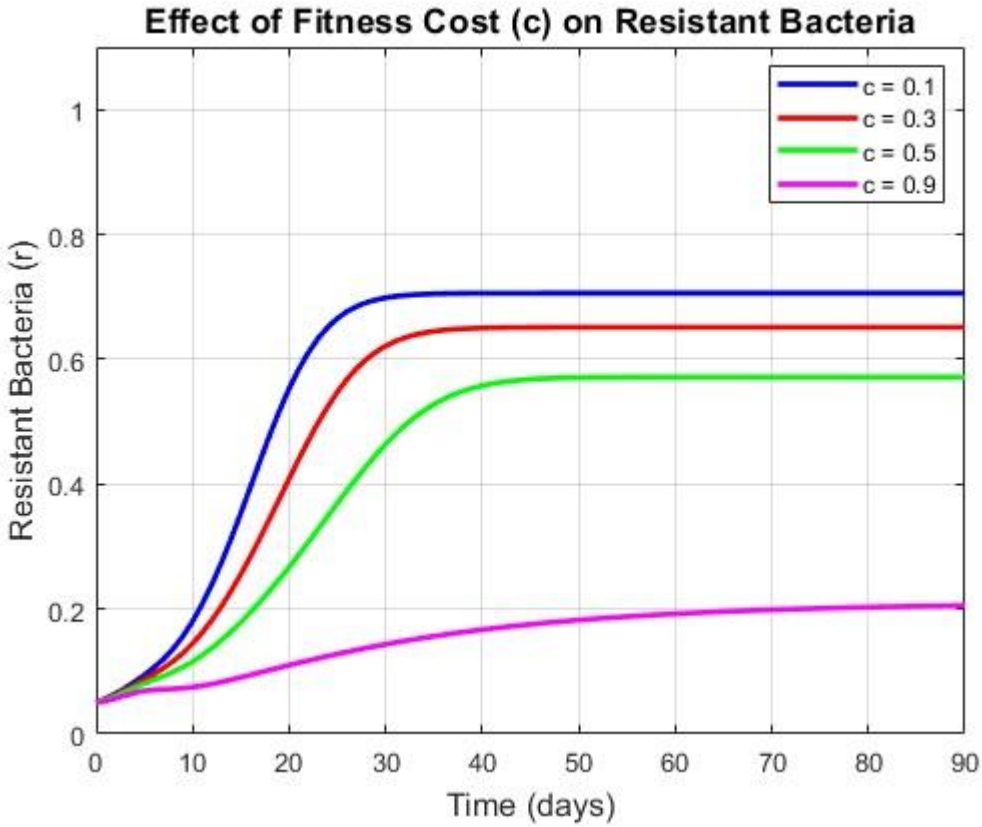
6.5 Effect of Immune System Strength



The plot shows total bacterial load over 90 days for four values of η (0.1, 0.3, 0.6, 1.0), where η represents the rate at which immune cells kill bacteria.

When η is low (0.1), the immune system is weak and bacterial load remains high throughout the treatment. As η increases, bacteria are cleared faster and settle at a lower equilibrium. At η = 1.0, the bacterial burden is reduced most aggressively. This confirms what the model predicts — a stronger immune response directly lowers the equilibrium bacterial load. In clinical terms, immunocompromised patients (low η) will carry a higher bacterial burden even under the same antibiotic treatment, making recovery harder.

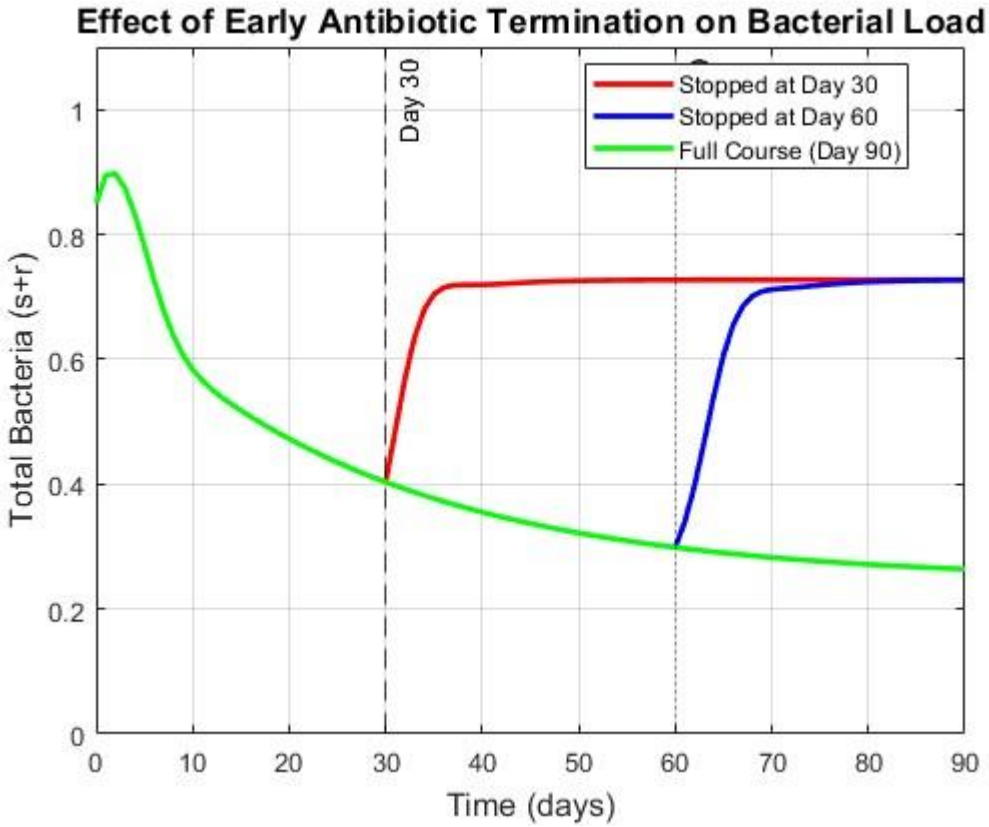
6.6 Effect of Fitness Cost



The plot shows how the resistant bacterial population evolves over 90 days for four values of fitness cost c (0.1, 0.3, 0.5, 0.9), where c directly reduces the growth rate of resistant bacteria as $\beta_R = (1-c)\beta_S$.

When c is small (0.1), resistant bacteria carry almost no fitness penalty and grow rapidly to a high equilibrium level. As c increases, resistant bacteria grow more slowly and stabilize at progressively lower levels. At c = 0.9, the fitness cost is severe and resistant bacteria remain at a very low level throughout. This shows that the biological cost of resistance is a natural limiting factor — the higher the cost, the less dominant resistant strains become. In clinical terms, this suggests that reducing antibiotic pressure can allow high-cost resistant strains to die out naturally over time.

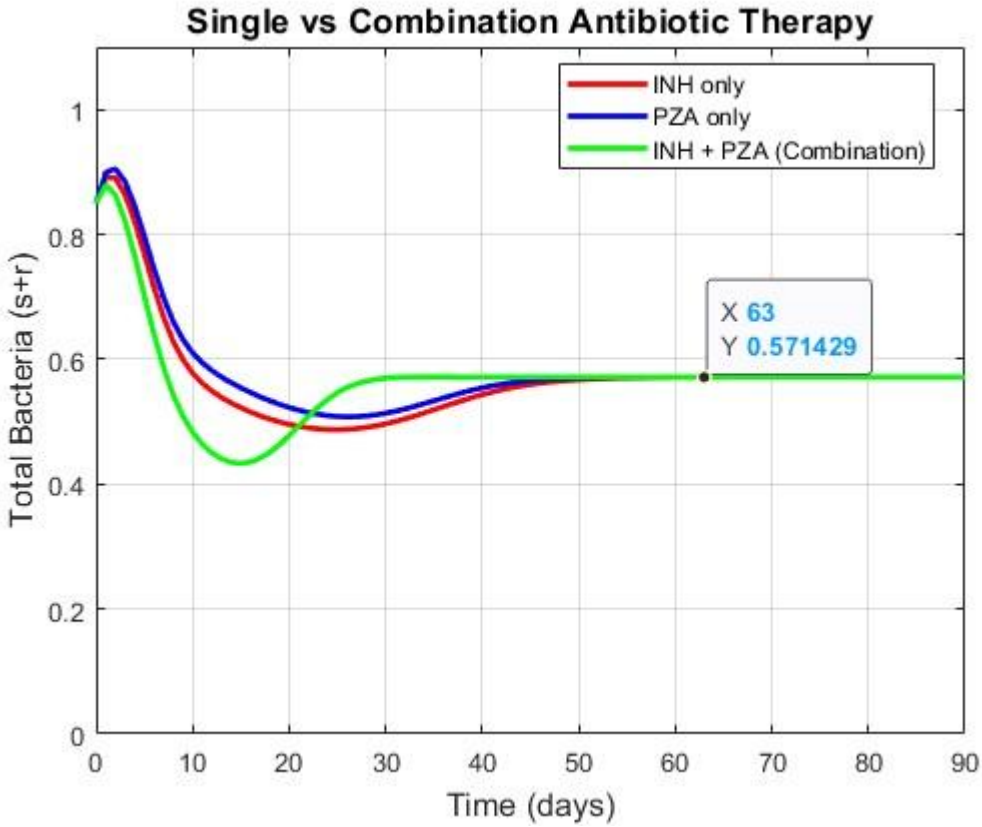
6.7 Early Antibiotic Termination



The plot compares total bacterial load when antibiotics are stopped at day 30, day 60, or completed for the full 90-day course.

When treatment is stopped at day 30, bacteria rebound sharply after the antibiotics are removed. The immune system alone is insufficient to control the infection and bacterial load climbs back toward high levels. Stopping at day 60 shows a partial rebound — bacteria had been suppressed further before treatment ended, so the rise is slower and settles at a higher level than the full course but lower than the day 30 case. Completing the full 90-day course produces the lowest final bacterial burden, with the system held at the E_2 equilibrium throughout. This result has a direct clinical message — incomplete antibiotic courses not only fail to clear the infection but actively create conditions for bacterial resurgence, and in practice, for resistance amplification.

6.8 Single vs Combination Therapy



The plot compares total bacterial load over 90 days under three treatment strategies: INH alone, PZA alone, and both antibiotics together.

Both single-drug treatments reduce bacterial load but stabilize at a higher equilibrium compared to combination therapy. INH alone performs better than PZA alone due to its stronger killing rate (d1 > d2), but neither single drug suppresses bacteria as effectively as the combination. The combination therapy produces the lowest final bacterial burden because each drug contributes independently to killing sensitive bacteria and slowing mutation, and their effects are additive in the model. This justifies the WHO recommendation of using multiple antibiotics simultaneously for infections such as tuberculosis — combination therapy reduces both the total bacterial load and the risk of

resistance emerging, since a mutation conferring resistance to one drug is unlikely to simultaneously confer resistance to both.

6.9 Sensitivity Analysis

Base A = 0.2000, Base B = 0.5714

Parameter	A_base	A_plus20	A_minus20	B_base	B_plus20	B_minus20
{'beta_S'}	0.2	0.3016	0.0638	0.5714	0.5714	0.5714
{'beta_R'}	0.2	0.2	0.2	0.5714	0.6154	0.5161
{'eta' }	0.2	0.1897	0.2115	0.5714	0.5263	0.625
{'alpha1' }	0.2	0.1964	0.2036	0.5714	0.5714	0.5714
{'alpha2' }	0.2	0.1891	0.2109	0.5714	0.5714	0.5714
{'d1' }	0.2	0.1727	0.2273	0.5714	0.5714	0.5714
{'d2' }	0.2	0.1364	0.2636	0.5714	0.5714	0.5714

The table shows how A and B change when each parameter is increased or decreased by 20% from its base value, while all other parameters are held constant.

A is most sensitive to β_S, η, d1, and d2. Increasing β_S raises A significantly because sensitive bacteria grow faster relative to immune and antibiotic pressure. Increasing η or the drug killing rates d1 and d2 lowers A, meaning

stronger immunity or stronger antibiotics push the system further toward clearing the infection. The mutation rates α1 and α2 have a smaller but notable effect on A since they contribute to removing sensitive bacteria.

B depends only on β_R and η. Increasing β_R raises B while increasing η lowers it. All drug-related parameters (d1, d2, α1, α2) have zero effect on B because resistant bacteria are not killed by antibiotics in this model — only the immune system acts on them.

The most critical parameters overall are **β_S, η, d1, and d2** — these four directly control whether the condition A < B or A > B holds, and therefore determine which equilibrium the system converges to. In clinical terms, this means immune strength and antibiotic killing efficacy are the two most important factors governing treatment outcome.

6.10 Summary of Results

Section	Parameters	Equilibrium Outcome	Biological Conclusion
6.3 Case I (A < B)	A = 0.20, B = 0.57	E2 = (0, 0.57, 0.57, 1, 1)	Sensitive bacteria eliminated. Resistant bacteria and immune cells coexist.
6.4 Case II (A > B)	A = 0.33, B = 0.25	E3 = (0.33, 0.00, 0.33, 1, 1)	All three populations persist. Infection not cleared.
6.5 Immune Strength	η: 0.1 → 1.0	Lower equilibrium as η increases	Stronger immune system reduces bacterial burden significantly.

| 6.6 Fitness Cost | c: 0.1 → 0.9 | Resistant bacteria decrease as c increases | High fitness cost limits resistant strain dominance. |

| 6.7 Early Termination | Stop at day 30, 60, 90 | Rebound ↑ if treatment stops early | Full course essential. Early stopping causes bacterial resurgence. |

| 6.8 Combination Therapy | INH only, PZA only, Both | Combination clears sensitive bacteria fastest | Two drugs together outperform either drug alone. |

| 6.9 Sensitivity | Each parameter ±20% | β_S, η, d1, d2 most critical | Drug killing rate & immune strength are key clinical levers. |

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The simulations carried out in this chapter collectively demonstrate that the mathematical model of Daşbaşı and Öztürk (2016) produces biologically meaningful and clinically relevant results across all scenarios tested. In Case I, where A < B, the system converges to equilibrium E_2 and sensitive bacteria are fully eliminated while resistant bacteria persist alongside immune cells. In Case II, where A > B, the system converges to E_3 and all three populations coexist indefinitely, representing a more severe and harder to treat infection state. The threshold condition A < B versus A > B therefore acts as the single most important determinant of treatment outcome in this model.

The parametric studies in sections 6.5 through 6.8 reveal how the system behaves under different clinical conditions. A stronger immune response (higher η) consistently lowers the bacterial burden at equilibrium, confirming that immunocompromised patients face worse outcomes under identical treatment. A higher fitness cost of resistance (higher c) naturally suppresses resistant strains, suggesting that antibiotic cycling strategies that allow resistance costs to accumulate could be beneficial. Early termination of antibiotics leads to clear bacterial rebound, providing mathematical justification for completing full treatment courses. Combination therapy

outperforms single drug treatment by driving sensitive bacteria to lower levels faster and reducing the conditions that favour resistance emergence.

The sensitivity analysis in section 6.9 identifies β_S, η, d_1, and d_2 as the most influential parameters in the model. These correspond to the bacterial growth rate, immune killing rate, and the bactericidal efficacy of each antibiotic. Small changes in these parameters can shift the system from the $A < B$ regime to $A > B$, completely changing the long term outcome. This highlights that in practice, even modest improvements in immune function or drug efficacy can be the difference between a controlled infection and one that persists indefinitely.